



Global Light Curve Attitude Estimation with Noisy Measurements and Inertia Uncertainty

Liam Robinson¹ · Carolin Frueh¹

Received: 26 January 2025 / Accepted: 19 November 2025 / Published online: 21 January 2026
© The Author(s) 2025

Abstract

Attitude information about human-made space objects can be an integral part of determining the status of individual satellites. If resolved optical images are unavailable due to distance or atmospheric effects, attitude information is still present in the time history of the observed object's brightness—its light curve. Extracting this information—a process known as attitude inversion—is plagued by many ambiguities, namely, the fact that infinitely many orientations may correspond to the same observed brightness value. In this work, we present a method for attitude inversion from the light curve and observation geometry when the object's shape and optical material properties are known but measurements are corrupted by noise from the celestial background, the image sensor, atmospheric turbulence, and the shot noise inherent to the object signal. We efficiently search the entire attitude space to yield collections of solutions that inherently account for ambiguities due to noise and symmetries in the object's geometry. This new optimization framework allows us to account for uncertainties in the object's moments of inertia directly. The capabilities and limitations of our approach are presented using synthetic light curves with low and high prior uncertainty in the object's inertia properties. To improve realism, all presented results include a model mismatch between the measured ground truth and the shape model used for inversion.

Keywords Light curve inversion · Attitude determination · Inverse problems · Space debris · Space situational awareness

✉ Liam Robinson
robin502@purdue.edu

Carolin Frueh
cfrueh@purdue.edu

¹ School of Aeronautics and Astronautics, Purdue University, 610 Purdue Mall, West Lafayette, IN 47906, USA

1 Introduction

Estimating attitude information for human-made space objects is critical for many space situational awareness (SSA) tasks. The orientation of an active satellite provides information about its mission objectives, while the angular velocity of space debris is necessary for planning and executing active debris removal [1]. Attitude information also plays a critical role in understanding the orbital perturbations caused by non-conservative forces like solar radiation pressure on high area-to-mass ratio objects [2] and inactive satellites [3]. Active satellites also require robust attitude estimation strategies in the event of on-orbit anomalies resulting in loss of attitude control and communication like that experienced by the CAPSTONE spacecraft in 2022 [4].

For objects in low-Earth orbit (LEO), ground-based radar observations may be used—under the right conditions—to estimate orientation and angular velocity for known objects using synthetic aperture techniques [5]. Even when resolved imagery is available for LEO objects, attitude [6] and shape estimation [7] are not straightforward. Above LEO, passive optical telescopes become more efficient for space surveillance as the power required for radar grows exponentially with altitude [8]. Because of atmospheric turbulence, diffraction, and the absolute distance between the observer and the space object, it is usually impossible to resolve an image of objects past LEO [9].

The measured brightness of the object in each optical image frame is the result of the observation geometry, as well as the object's attitude [10], shape [11, 12], and reflective properties of its surfaces [13]. The observation geometry may be readily determined through orbit determination, but separating the remaining effects is difficult without additional information as infinitely many combinations of geometry, materials, attitude, and inertia properties may produce an identical brightness value [14]. In this way, inverting the attitude from a single brightness measurement is generally ambiguous. The time history of brightness variations for a particular object is termed the light curve, and the extraction of any information from the light curve is known as light curve inversion.

Light curve inversion for attitude information is one example of a wider field of inverse problems whose characteristics have been studied at great length. In the most general case, the solution to any inverse problem may be expressed as a probability distribution over the space of model parameters [15]. Given uncertainty in the measurements and in the modeling assumptions, we generally cannot identify a singular solution, and must attempt to identify or parameterize the space of solutions that are equally effective at reproducing the measurements [15, 16].

Many inversion methods like multiple-model adaptive estimation (MMAE) [17, 18] and unscented or extended Kalman filters [19–21] have been developed for cases when measurements are corrupted by noise but a good attitude state guess is available. These filters are sensitive to the quality of the initial guess; they will not converge if the initial guess is far from the truth. Different filtering approaches have noted divergence for initial guess attitude errors above approximately 40° [18], 30° [17], and 5° [20]. Similar convergence bounds on the initial angular velocity error as small as 0.5 [18], 0.17 [21], and 0.03 [20] degrees per second have been observed.

When no reliable initial guess is available, the attitude inversion problem must be solved globally as a more general inverse problem. This is the case we are concerned with. The particular method chosen to solve this global inversion problem is merely a choice of search method to approximate the probability distribution of models given the observations. Particle filters accomplish this by beginning from an initial solution distribution and iteratively updating particle positions to approximate a probability density of the unknown attitude parameters [22, 23]. While particle filters may naturally account for ambiguities in the problem, their formulation can be complex, requiring special treatment of the non-additive attitude variables [22]. Probability hypothesis density or other Bayesian multi-hypothesis filters [19, 24] require a prior distribution for the attitude state, which is challenging to define meaningfully for an arbitrary space object.

Other global attitude inversion methods include simulated annealing [20, 25], genetic algorithms [20, 25, 26], and particle swarm optimizers (PSOs) [14, 20, 25, 27, 28]. These methods often possess numerous tunable hyperparameters that must be optimized on a case-by-case basis [25]. These approaches are inherently stochastic, making it difficult to predict their behavior more generally.

Independent of the selected inversion algorithm, past works often make unrealistic assumptions during light curve simulation and inversion in the treatment of three major areas: measurement noise, self-shadowing, and shape model accuracy.

As the brightness of an object and the observation geometry change, different physical processes dominate the photometric measurement noise [29]. Nevertheless, noise is often ignored in the inversion procedure. If no noise is added in light curve simulation, solvers may achieve high degrees of convergence at the cost of realism [14, 19, 28]. Although the Bayesian filter-based methods naturally account for the measurement noise in the estimation process, as mentioned before, the particular implementations of PSOs, genetic algorithms, and simulated annealing techniques presented in prior works do not. Methods that do not respect the possibly time-varying uncertainty in each observation are susceptible to overfitting noisy data, producing attitude profiles that do not match the ground truth, although they fit the measurements well. Past works that do add measurement noise on the light curve do so by approximating it as a fixed signal-to-noise ratio (SNR) or fixed standard deviation Gaussian process [17, 22–24, 26, 30, 31], conditions which are not generally satisfied by real observations. Among works that use Gaussian noise, the noise is often added to observations in apparent magnitude, introducing a non-physical bias in the synthetic measurements.

Almost all human-made space objects have some degree of non-convex geometry, leading to self-shadowing effects that are difficult to compute efficiently. Despite this, many works only present results for convex objects [17, 22, 31, 32] that lack self-shadowing effects entirely. Results that do use non-convex objects often neglect self-shadowing effects due to computational complexity and simulate the objects as if they were convex [25–27, 33]. Ignoring the reality of self-shadowing limits the applicability of a method and the generalizability of the presented results. Select recent publications do account for self-shadowing through face-level approximations [11, 30, 34], pixel-level approximations [12, 27, 35], and neural networks [14, 28]. Semi-

analytic self-shadowing methods exist and have been used for light curve simulation [20, 36], but not during the inversion process due to computation time concerns.

In reality, the shape of the object being observed often must be estimated from pre-launch imagery or 3D renderings. Due to time on orbit, simplifications in renderings, or difficulty in estimating material reflection coefficients, there will always be a gap between the object being observed and the shape assumed during inversion. As a result, a novel attitude inversion algorithm should be tested with a similar model mismatch.

This work extends recent literature on particle swarm methods by combining the advantages of PSOs—namely that no initial guess is needed and they inherently find ambiguous solutions—with new improvements to address their weaknesses. We present an approach to solving the attitude inversion problem via direct nonlinear optimization that simultaneously simplifies the inversion algorithm while improving result realism. We account for uncertainty in the object’s mass properties by directly optimizing the moment of inertia given a user-supplied prior distribution. Our approach features a fast, semi-analytic shadowing algorithm and accounts for realistic measurement noise processes both in simulation and inversion. The primary contribution of this work is closing the gap between simulation and the reality of real observations. By relaxing the measurement noise, self-shadowing, and known inertia assumptions made by other methods, we set the stage for reliable full-state light curve attitude inversion on real data.

This paper proceeds by presenting our solver in Sect. 2 and formulating our chosen measurement model in Sect. 3. Section 4 details results for both low and high uncertainty in the moments of inertia are presented for high-fidelity synthetic data on a convex ATLAS V rocket body and a non-convex HYLAS box-wing satellite.

2 Attitude Inversion Method

2.1 Light Curve Loss Functions

Many non-iterative methods presented in past works [14, 27, 28] choose a ℓ_2 loss function—or the closely related RMSE loss function—to express the error in the estimated light curve $\hat{\mathbf{S}} = [\hat{S}_1, \hat{S}_2, \dots, \hat{S}_m]^T$ compared to the observations $\mathbf{S} = [S_1, S_2, \dots, S_m]^T$:

$$f_{\ell_2}(\hat{\mathbf{S}}, \mathbf{S}) = \sum_{k=1}^m \left(\hat{S}_k - S_k \right)^2. \quad (1)$$

This loss function is known to cause problems as it attributes much larger errors to brighter regions of the light curve, leading to solutions that overfit bright specular glints while ignoring smaller variations in the signal. The ℓ_2 loss also ignores the noise characteristics at each timestep as it is blind to variations in the measurement variance. For a light curve of length m corrupted by zero-mean Gaussian measure-

ment noise with variance σ_k^2 at timestep k , the likelihood of the entire light curve is the product of each timestep's probability density function:

$$L = \prod_{k=1}^m \left[\frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(\hat{S}_k - S_k)^2}{2\sigma_k^2}} \right]. \quad (2)$$

Minimizing $-\ln L$ maximizes the likelihood L while transforming the product into a sum, yielding the final negative log-likelihood loss function f_{NLL} by dropping the constant term and normalizing by the number of data points m :

$$f_{\text{NLL}}(\hat{\mathbf{S}}, \mathbf{S}) = \frac{1}{m} \sum_{k=1}^m \left[\ln \sigma_k + \frac{1}{2} \left(\frac{S_k - \hat{S}_k}{\sigma_k} \right)^2 \right]. \quad (3)$$

Optionally, the estimated mean light curve $\hat{\mathbf{S}}$ can be rescaled by the factor $\|\mathbf{S}\| / \|\hat{\mathbf{S}}\|$ before computing the loss, where $\|\cdot\|$ denotes the ℓ_2 norm of the input vector, forcing both signals to have equal magnitude. It must be noted that the standard deviations σ_k do not need to be renormalized as they originate from the observations and are not a function of the estimated mean light curve. This renormalization helps address model mismatch by correcting systematic errors in albedo-area, allowing light curves of the correct shape but incorrect scale to have a low loss. Better convergence is usually achieved using this normalization; all results presented in this work were produced using it.

2.2 State Definition

The state vector $\mathbf{x}$ is the concatenation of the Modified Rodriguez Parameter (MRP) as the kinematic attitude variable [37], the body frame angular velocity, and the inertia tensor ratios:

$$\mathbf{x} = \begin{bmatrix} p_1 & p_2 & p_3 & \omega_1 & \omega_2 & \omega_3 & \frac{J_y}{J_x} & \frac{J_z}{J_x} \end{bmatrix}^T. \quad (4)$$

The principal axes frame is body frame used throughout this work, with principal moments of inertia (PMOIs) J_x , J_y , and J_z :

$$\mathbf{J} = \begin{bmatrix} J_x & 0 & 0 \\ 0 & J_y & 0 \\ 0 & 0 & J_z \end{bmatrix}. \quad (5)$$

The dynamics of the target object are well-approximated as torque-free over the course of a single light curve—usually on the scale of a few minutes—for all orbital regimes apart from high area-to-mass ratio objects in very low-Earth orbits, where drag is a significant torque. Precisely defining the altitude and area-to-mass ratio that constitutes a violation of the torque-free assumption is a function of the amount

of information available about the object prior to inversion; lower uncertainty often enables a more stringent constraint. Instead of including all three PMOIs in the state definition, we save one dimension by only defining the ratios J_y/J_x and J_z/J_x as states, taking advantage of the fact that the differential equations governing torque-free motion are only a function of the inertia ratios. Note that in this formulation, it is always true that $J_x = 1$.

2.3 Initial Condition Sampling

Given that we assume no prior attitude knowledge, we use a uniform prior for the angular velocity and orientation. For the orientation component of the initial state, we first sample uniform random quaternions via [38, 39]:

$$\hat{q} = \frac{1}{\sqrt{g_1^2 + g_2^2 + g_3^2 + g_4^2}} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{bmatrix}, \quad (6)$$

where g_1, g_2, g_3 , and g_4 are independent and identically distributed samples from the standard Gaussian distribution $\mathcal{N}(0, 1)$. Sampling independent Gaussian variables draws from a spherically symmetric distribution in any number of dimensions n . By normalizing, we are precisely drawing from a four-dimensional distribution with an identity covariance, producing uniform orientations as the quaternions have a uniform density over the $SO(3)$ [40].

This quaternion $\hat{q} = [q_1, q_2, q_3, q_4]^T$ —defined with q_4 as the scalar component—is converted into an MRP p before insertion into the state vector [41]:

$$p = \frac{1}{q_4 + 1} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}. \quad (7)$$

The angular velocity is sampled uniformly in the ball of radius $\omega_{\max}$ using the same algorithm as the quaternion, scaled by $r^{1/3}$ to account for the scaling of the volume of an infinitesimally thin spherical shell with radius r where $r \sim \mathcal{U}(0, 1)$:

$$\omega = \frac{\omega_{\max} r^{1/3}}{\sqrt{g_5^2 + g_6^2 + g_7^2}} \begin{bmatrix} g_5 \\ g_6 \\ g_7 \end{bmatrix}, \quad (8)$$

where g_5, g_6 , and g_7 are samples from the standard Gaussian distribution $\mathcal{N}(0, 1)$ to distribute the ω samples uniformly on the sphere.

The inertia tensor ratios J_y/J_x and J_z/J_x are sampled from prior distributions defined arbitrarily based on the information available about the observed targets. In this work, we use a Gaussian distribution centered on the truth value of these ratios with a standard deviation defined by σ_J .

2.4 Solver Selection

The light curve inversion problem is challenging to optimize efficiently due to its expensive loss function. Computing the loss of a particular initial attitude state requires propagating the rigid body dynamics forward to all the observation epochs and computing the expected light curve value at each time to find the brightness error. If the object's shape is simple—composed of maximally a few dozen faces—the bottleneck of the loss function will be the attitude integration. If the object is modeled with more detail or is highly non-convex, the computation of the brightness values will be the limiting factor. The Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm [42–45] was selected due to its relative ease of implementation and its history of success for medium-scale optimization problems in the context of particle-based non-convex optimization algorithms [46].

If no initial guess is available, the goal of any global search method—be it BFGS, simulated annealing, particle swarm, genetic algorithm, particle filter, or a multi-hypothesis filter—is to map the solution space by finding any and all state vectors that predict the observations with high likelihood. While many of these solvers are capable of exhaustively searching the space efficiently, the objective function must be posed thoughtfully to fully capture the inherent ambiguities of the attitude inversion problem. As a result, we focus much of this work on improving the realism of the forward light curve model.

2.5 Solver Execution

From a given initial condition x_0 sampled using the procedure in Sect. 2.3, the BFGS algorithm iteratively updates that guess to search for a nearby local minimum in the loss function defined by Eq. (3). Each update is performed by propagating the initial attitude state to each measurement timestep, and computing a mean light curve value at each. These mean light curve values are combined with the observed brightness values and their approximate standard deviations to yield a loss function value. Once the loss value at the current state position is known, a two-point finite difference method approximates the gradient. A BFGS Hessian update is then applied, which is used to determine a step direction and magnitude that satisfies the Wolfe conditions [47]. The BFGS Hessian update is iterative and low-rank, requiring only gradient information at the current iteration. As a result, the Hessian update requires k function evaluations at each iteration, where k is the dimension of the state space [42]. Convergence is reached when either the gradient norm is below a given tolerance, the loss function is below a given tolerance, a maximum number of iterations is reached, or a maximum number of loss function evaluations is exceeded. This process is repeated in parallel for n total initial conditions, yielding n optimized initial state guesses. These optimized states can then be ranked by their loss value for post-processing. We implemented the attitude inversion algorithm presented in this work in the Taichi language [48], making it fully multithreaded and allowing it to be compiled to the CPU or GPU for execution.

2.5.1 Classifying Converged Solutions

Due to model mismatch, it is nontrivial to determine the quality of an inversion result when the ground truth is unavailable. In general, it is impossible to determine *a priori* how close the assumed model can get to the measured light curve. With a large enough model mismatch, the assumed model may be unable to reproduce any of the measurements in any attitude profile meaningfully. Alternatively, it may be able to overfit and produce the measurements exactly. For these reasons, converged initial guesses with unusually low loss values should be treated with extreme skepticism, while high loss values past a defined limit can be discarded entirely.

We roughly classify solutions that overfit the data by computing the loss via Eq. (3) associated with placing the assumed model in the truth attitude profile. This amounts to comparing the mean light curve of the observed model with the simulated noisy measurements. This loss value is denoted f_{best} , and we denote solutions as converged if $f_i < \alpha f_{\text{best}}$ where $\alpha > 1$, and overfit if $f_i < f_{\text{best}}$.

If no ground truth is available, f_{best} can simply be taken to be the lowest error solution. A value for α should be chosen based on user's tolerance for error in the light curve based on the available prior information on the space object being studied and the measurement noise. If there is high confidence in the object's geometry and materials, a lower value of α should be used to only select the best solutions. If little is known about the object, there is a higher likelihood that an attitude profile may be close to the truth although it does not fit the light curve well, therefore implying a higher bound α . In practice, this is an iterative process. If the first chosen α permits light curves that are too far from the observations, a higher value should be chosen. Likewise, if only a small number of similar solutions are available for a given α , the bound should be raised to discourage over-convergence.

3 Light Curve Measurement Prediction Model

The inversion method in Sect. 2 is designed to operate independently of the particular algorithm chosen to compute the light curve and its standard deviation in the prediction step of the solver. This section details one particular set of choices for the object dynamics, shape representation, and light curve computation that are optimized for speed and reliability for use in a global attitude inversion method.

3.1 Attitude State Representation

We choose the MRP to describe the time evolution of the initial attitude state of the space object in inertial space. The MRP $\mathbf{p} = [p_1, p_2, p_3]^T$ is easily mapped to the direction cosine matrix (DCM) C via [41]:

$$C(\mathbf{p}) = I_3 + \frac{8\tilde{\mathbf{p}}\tilde{\mathbf{p}} - 4\left(1 - \|\mathbf{p}\|^2\right)\tilde{\mathbf{p}}}{\left(1 + \|\mathbf{p}\|^2\right)^2}, \quad (9)$$

where I_3 is the 3×3 identity matrix, $\|\mathbf{p}\|^2 = p_1^2 + p_2^2 + p_3^2$, and $\tilde{\mathbf{v}}$ is the cross product matrix of vector $\mathbf{v}$ such that $\mathbf{v} \times \mathbf{b} = \tilde{\mathbf{v}}\mathbf{b}$ for another vector $\mathbf{b} \in \mathbb{R}^3$. A vector $\mathbf{v}$ in inertial space is mapped to the body frame via $C(\mathbf{p})\mathbf{v}$.

The kinematic and dynamic differential equations are given as a function of the $[3 \times 3]$ symmetric inertia tensor J and a possible external torque vector $\mathbf{M}$ by [41]:

$$\dot{\mathbf{p}} = \frac{1}{4} [(1 - \mathbf{p} \cdot \mathbf{p}) + 2\mathbf{p} \times \boldsymbol{\omega} + 2(\boldsymbol{\omega} \cdot \mathbf{p})\mathbf{p}] \quad (10)$$

$$\dot{\boldsymbol{\omega}} = J^{-1} [(J\boldsymbol{\omega}) \times \boldsymbol{\omega} + \mathbf{M}] \quad (11)$$

Given initial conditions $\mathbf{p}(0)$ and $\boldsymbol{\omega}(0)$, we numerically integrate Eqs. (10) and (11) in the body axes using the fourth-order Runge–Kutta algorithm (RK4)—the so-called famous method listed in [49]—to yield the orientation at each measurement time for light curve computation.

To measure the angular error between two orientations $\mathbf{p}$ and $\mathbf{p}'$, we find the relative orientation $\mathbf{p}''$ [41]:

$$\mathbf{p}'' = \frac{(1 - \|\mathbf{p}'\|^2)\mathbf{p} - (1 - \|\mathbf{p}\|^2)\mathbf{p}' - 2\mathbf{p} \times \mathbf{p}'}{1 + \|\mathbf{p}'\|^2\|\mathbf{p}\|^2 + 2\mathbf{p}' \cdot \mathbf{p}}, \quad (12)$$

such that the principal rotation angle θ between $\mathbf{p}$ and $\mathbf{p}'$ is simply [41]:

$$\theta = 4 \tan^{-1} (\|\mathbf{p}''\|). \quad (13)$$

Some space objects, like rocket bodies and spin-stabilized satellites, are well-approximated as having axisymmetric inertia tensors with two equal principal moments of inertia J_t and one distinct axial moment of inertia J_a . Their rigid body motion is characterized by the sum of two rotations—a precession rate about the inertial angular momentum vector and a spin rate about the axis corresponding to J_a [50]. Given an initial body-frame angular velocity $\boldsymbol{\omega}(0)$, the spin and precession rates, notated ω_s and ω_p , respectively, are given by [51]:

$$\omega_p = \frac{\|\boldsymbol{\omega}(0)J\|}{J_t} \quad (14)$$

$$\omega_s = \frac{J_t - J_a}{J_t} \omega_a(0), \quad (15)$$

where $\omega_a(0)$ is the initial angular velocity about the axis corresponding to J_a .

3.2 Shape Representation

Human-made space objects may be well-approximated by triangulated meshes. We define the n vertices of an object by the set $\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and its m faces by a

collection of vertex sets with the i th element $\mathcal{F}_i = \{\mathbf{v}_{i,1}, \mathbf{v}_{i,2}, \mathbf{v}_{i,3}\}$. We must solve for each face's unit vector and area to compute light curves for an object. For each triangular face $\mathcal{F}_i$, the outward-pointing face normal $\hat{\mathbf{n}}_i$ is computed with [52]:

$$\hat{\mathbf{n}}_i = \frac{(\mathbf{v}_{i,2} - \mathbf{v}_{i,1}) \times (\mathbf{v}_{i,3} - \mathbf{v}_{i,1})}{\|(\mathbf{v}_{i,2} - \mathbf{v}_{i,1}) \times (\mathbf{v}_{i,3} - \mathbf{v}_{i,1})\|}. \quad (16)$$

The vertices of each face must be ordered counterclockwise to ensure that each face's normal vector is outward-pointing. The total area of the i th face is computed via $a_i = A(\mathcal{F}_i)$ using the equation for the area of an arbitrary planar convex polygon $\mathcal{P}$ with k counterclockwise vertices $\mathbf{v}_j$ [53]:

$$A(\mathcal{P}) = \sum_{j=1}^{k-2} \frac{\|(\mathbf{v}_{j+1} - \mathbf{v}_1) \times (\mathbf{v}_{j+2} - \mathbf{v}_1)\|}{2}. \quad (17)$$

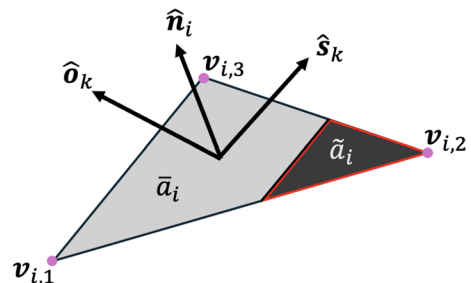
When the input face is triangular, the sum in Eq. (17) is unnecessary as $k = 3$.

Figure 1 displays the relevant geometry for this computation on one triangular face, including the body-frame unit vectors $\hat{\mathbf{o}}_k$ and $\hat{\mathbf{s}}_k$ at the k th timestep pointing from the center of mass of the object to the observer and Sun, respectively. Depending on the object's geometry, some faces may be able to cast shadows onto other parts of the geometry. To account for this self-shadowing, the total area a_i is subdivided into the unshadowed area $\bar{a}_i$ and the shadowed area $\tilde{a}_i$. We now present an semi-analytic approach to computing these areas for non-convex objects.

3.3 Semi-Analytic Self-Shadowing

In this work, we present an algorithm for computing semi-analytic self-shadowing boundaries, dramatically accelerating the light curve computation for simple non-convex objects. Pixel-wise self-shadowing is still used for noisy light curve generation, owing to its efficiency when dealing with the high-fidelity 3D models used for the ground truth, but all inversion results are produced using the new semi-analytic shadowing procedure. This core of the semi-analytic shadowing algorithm is summarized graphically in Fig. 2, where faces shadowing a subject polygon are projected into its plane before being clipped against the subject polygon and their mutual higher-order intersections. The areas of the resulting intersections are used to com-

Fig. 1 Facet geometry including the observer direction $\hat{\mathbf{o}}$, Sun direction $\hat{\mathbf{s}}$, normal vector $\hat{\mathbf{n}}_i$, unshadowed area $\bar{a}_i$, shadowed area $\tilde{a}_i$, and counterclockwise vertex locations $\{\mathbf{v}_{i,1}, \mathbf{v}_{i,2}, \mathbf{v}_{i,3}\}$



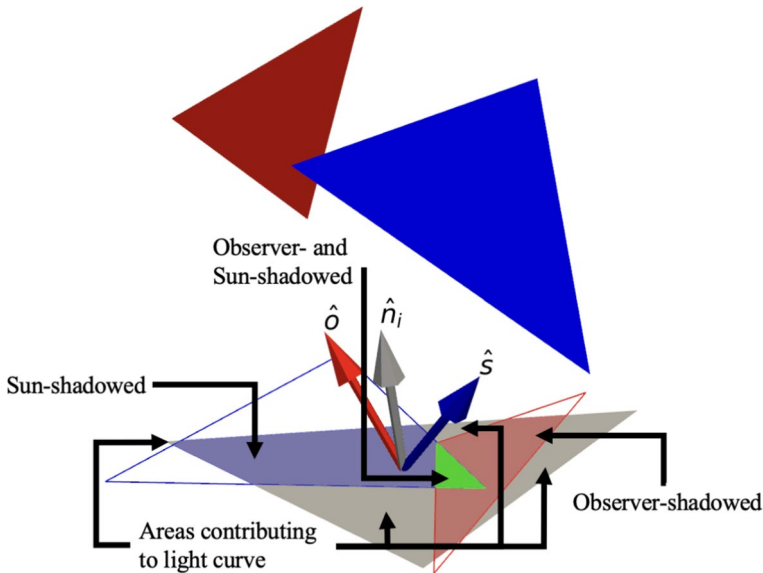


Fig. 2 Semi-analytic shadowing example for simultaneous observer and Sun self-shadowing. The face in red is shadowing the subject polygon from the observer’s perspective, likewise for the face in blue from the Sun’s perspective. Shadowing faces are projected into the plane of the subject polygon before being clipped and intersected with each other to compute the remaining area

pute the total shadowed area $\tilde{a}_i$. We begin this by projecting the candidate shadowing polygons onto the subject polygon.

3.3.1 Projecting Shadowing Polygons onto the Subject Polygon

Each polygon—e.g., the triangle in Fig. 1—with n edges is arranged as a list of points in counterclockwise order where the $i \in [0, n]$ and $(i + 1) \pmod{n}$ -th elements form an edge. Given a vertex on the subject polygon $\mathbf{v}_p$ and the subject’s normal vector $\hat{\mathbf{n}}_p$, the perpendicular distance from the subject polygon to a point on the shadowing polygon $\mathbf{v}_s$ is given by:

$$d_{\perp} = (\mathbf{v}_s - \mathbf{v}_p) \cdot \hat{\mathbf{n}}_p. \quad (18)$$

The direction of projection—i.e., the direction the shadow is cast from, defined as $\hat{\mathbf{p}} \in \{\hat{\mathbf{o}}, \hat{\mathbf{s}}\}$ to capture the possible Sun and observer shadowing conditions—is not generally perpendicular to the subject’s normal vector. As a result, we must translate $\mathbf{v}_s$ by $d_{\perp}/(\hat{\mathbf{n}}_p \cdot \hat{\mathbf{p}})$ to reach the plane of the subject. This yields the projection of $\mathbf{v}_s$:

$$\text{proj}_{\mathcal{P}_p} \mathbf{v}_s = \mathbf{v}_s - \left(\frac{d_{\perp}}{\hat{\mathbf{n}}_p \cdot \hat{\mathbf{p}}} \right) \hat{\mathbf{p}} \quad (19)$$

Repeating this process for each shadowing polygon vertex yields a projected copy of the shadowing polygon coplanar with the subject polygon. Before computing the

intersection, it is advantageous to express the vertices in $\mathbb{R}^2$ by orthonormalizing a set of basis vectors $\{\hat{\mathbf{u}}_1, \hat{\mathbf{u}}_2\}$ in the plane via two points $\mathbf{v}_1$ and $\mathbf{v}_2$ on the subject polygon along with the subject's unit normal vector $\hat{\mathbf{n}}$:

$$\begin{aligned}\hat{\mathbf{u}}_1 &= \frac{\mathbf{v}_2 - \mathbf{v}_1}{\|\mathbf{v}_2 - \mathbf{v}_1\|} \\ \hat{\mathbf{u}}_2 &= \hat{\mathbf{n}} \times \hat{\mathbf{u}}_1.\end{aligned}\quad (20)$$

With these unit vectors, we find the planar vertices of the subject polygon $\mathcal{S}$ and projected shadowing polygon $\mathcal{C}$:

$$\mathcal{S}_p = \left\{ \begin{bmatrix} \mathbf{v} \cdot \hat{\mathbf{u}}_1 \\ \mathbf{v} \cdot \hat{\mathbf{u}}_2 \end{bmatrix} \mid \mathbf{v} \in \mathcal{S} \right\} \quad (21)$$

$$\mathcal{C}_p = \left\{ \begin{bmatrix} \mathbf{v} \cdot \hat{\mathbf{u}}_1 \\ \mathbf{v} \cdot \hat{\mathbf{u}}_2 \end{bmatrix} \mid \mathbf{v} \in \mathcal{C} \right\} \quad (22)$$

Given k shadowing polygons $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_k\}$ and a subject polygon $\mathcal{S}$, we aim to determine the union of the intersections of $\mathcal{S}_p$ with each shadowing polygon $\mathcal{C}_p$ such that:

$$\mathcal{S}_p^* = \bigcup_{i=1}^k \mathcal{S}_p \cap \mathcal{C}_{p,i}. \quad (23)$$

The polygon $\mathcal{S}_p^*$ in Eq. (23) represents the planar region of $\mathcal{S}_p$ shadowed by one or more of $\mathcal{C}_{p,i}$. In order to compute $\mathcal{S}_p^*$, we must develop an approach to compute the intersections and union in Eq. (23).

3.3.2 Polygon Intersections

To compute Eq. (23), we must choose methods for clipping two polygons and determining the union area shared between n shadows. To compute the intersecting area of a convex clipping polygon with a possibly non-convex subject polygon, we use the Sutherland-Hodgeman algorithm [54]. Sutherland-Hodgeman computes intersections by testing whether the start and end points of a subject polygon edge fall *outside* or *inside* a particular edge with endpoints $\mathbf{v}_1, \mathbf{v}_2$ of the clipping polygon [54].

The Sutherland-Hodgeman algorithm proceeds by iterating over all edges of the subject polygon for the first edge of the clipping polygon, building up a new polygon $\mathcal{S}'$, which replaces the original subject polygon on future iterations. The process of iterating over the edges of $\mathcal{S}'$ repeats for the remaining edges of $\mathcal{C}$, where afterward, we have the intersection of the input polygons such that $\mathcal{S} \cap \mathcal{C} = \mathcal{S}'$. This procedure is displayed in Fig. 3.

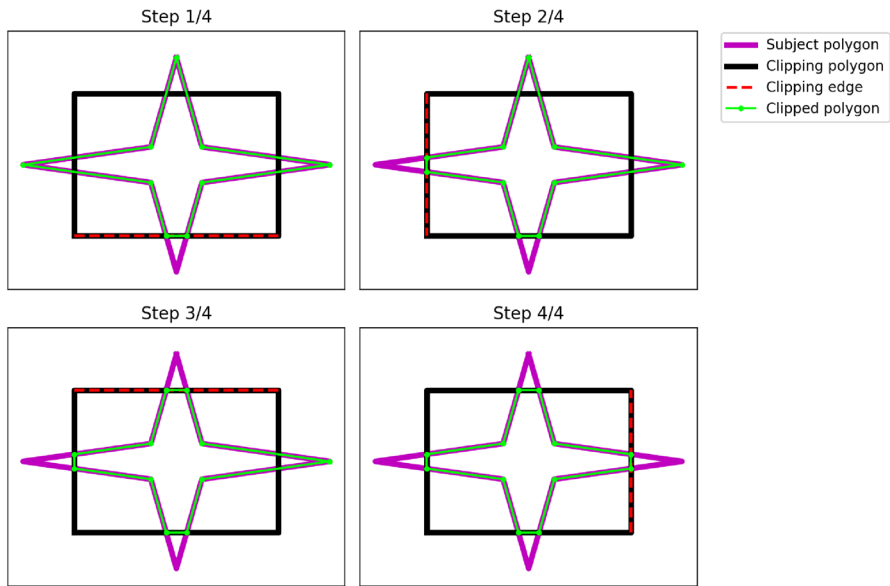


Fig. 3 Sutherland-Hodgeman polygon clipping algorithm results. The subject polygon is clipped against the selected clipping edge at each step. After iterating through all clipping edges, notice that all regions of the non-convex subject polygon outside the convex clipping area are removed

3.3.3 Computing the Unshadowed Area

To determine the total unshadowed area of a face, we project all faces into its plane twice—from the projection directions of the Sun and observer—and use the Sutherland-Hodgeman algorithm to compute the intersection polygon. Note that to compute the full union of intersections prescribed by Eq. (23), we must check for mutual higher-order intersections if multiple intersection polygons have a non-zero area. In principle, this recursion continues to an arbitrary depth until all possible polygon intersections are accounted for. In general, the unshadowed area $\bar{a}$ of the subject face with total area a shadowed by n other faces is given by:

$$\bar{a} = a - \sum_{d=1}^n \sum_{K \in \text{comb}(n,d)} (-1)^{d+1} A \left(\bigcap_{k \in K} \mathcal{P}_k \right). \quad (24)$$

Here, $\mathcal{P}_k$ is the intersection polygon of the k th shadowing face with the subject face, $A(\cdot)$ is the area function for a planar polygon defined by Eq. (17), and $\text{comb}(n, d)$ is the set of all combinations of d positive integers up to and including n . In practice, searching for intersections up to a maximum depth is often sufficient to limit computational complexity, this paper presents results for a maximum intersection depth of $d = 2$.

Notably, this formulation to compute the total unshadowed area does not solve for the geometry of the shadowed region. This is advantageous for certain shadowing geometries, such as a thin antenna casting a shadow onto a solar panel which cuts

its unshadowed region into two disjoint areas. We avoid this by only solving for the relevant areas of projected face intersections, which remain convex as long as the individual faces are convex.

3.3.4 The Shadow Cache

Determining the shadowing configuration, i.e., which faces are shadowing which other faces, is expensive. We can avoid significant repeated computation by taking advantage of the fact that many neighboring illumination directions result in the same shadowing configuration. Computing which faces self-shadow specific other faces over a grid of azimuth and elevation values in the object body frame yields a lookup table—the so-called ‘shadow cache’—which is accessed at runtime to speed up light curve evaluation. These azimuth and elevation values, denoted Az and El , respectively, are related to a body-frame unit vector $\hat{e}$ via:

$$\hat{e} = \begin{bmatrix} \cos(El) \cdot \cos(Az) \\ \cos(El) \cdot \sin(Az) \\ \sin(El) \end{bmatrix}. \quad (25)$$

For an azimuth sample count n_{Az} , an elevation sample count n_{El} , and an object face count of n_f , the shadow cache is a four-dimensional boolean array with the shape $n_{Az} \times n_{El} \times n_f \times n_f$. We have found in testing that n_{Az} and n_{El} counts of 200 are more than sufficient to capture all shadowing conditions. Truncating to the nearest cached value introduces errors at the boundaries where a face transitions from being shadowed to unshadowed. This causes negligible brightness error as the observation geometries triggering this truncation error involve faces that are either barely visible to the observer, or barely illuminated by the Sun. If this boundary error is of concern, a higher n_{Az} or n_{El} may be used to reduce this phenomenon. For a body-frame direction corresponding to index i, j and object polygon indices k and l , the (i, j, k, l) entry of the cache expresses whether face k shadows face l when viewed from the input direction. The body frame azimuth and elevation pair (Az, El) corresponding to cache indices (i, j) is computed via:

$$Az = \frac{2\pi i}{n_{Az}} \quad (26)$$

$$El = \frac{\pi j}{n_{El}} - \frac{\pi}{2}. \quad (27)$$

To compute each entry of the cache, we initialize the cache as False and project the face k onto face l along the direction defined by (i, j) . If a non-zero intersection area is found, we change the cache entry to True.

During attitude inversion, we compute the shadowing configuration for query directions not present in the cache by finding the nearest cached direction, rounding down in both azimuth and elevation in the first two dimensions of the cache instead of computing intersections for all n_f choose 2 face combinations. This rounding is

performed in this work via the floor operator $\lfloor \cdot \rfloor$, although traditional rounding may yield lower errors for low values of n_{Az} or n_{El} . We transform from an azimuth and elevation pair (Az, El) to cache indices (i, j) via:

$$i = \left\lfloor \frac{Az}{2\pi} \cdot (n_{Az} - 1) \right\rfloor \quad (28)$$

$$j = \left\lfloor \left(\frac{El}{\pi} + \frac{1}{2} \right) \cdot (n_{El} - 1) \right\rfloor. \quad (29)$$

After all shadowing combinations from the Sun and observer directions have been identified using the cache, Eq. (24) is used to determine the unshadowed area of each face, which is substituted into Eq. (33) to compute a light curve value. Figure 4 shows an example of one of these caches, with one of the cache entries highlighted and displayed on the right.

The shadow cache is memory efficient as each shadowing condition only requires one bit. For $n_{Az} = n_{El} = 200$ and $n_f = 100$, the cache will occupy 50 MB of space in memory.

3.4 The Bidirectional Reflectance Distribution Function

For each illuminated face of the object, we determine the amount of incoming light reflected in the observer's direction per steradian with the so-called bidirectional reflectance distribution function (BRDF). In this paper, we choose the Blinn-Phong BRDF to compute reflectance values as it is both simple and energy-conserving [55]. We choose a relatively simple BRDF—as opposed to microface models like Cook-Torrance [56], anisotropic models like Ashikhmin-Shirley [57], or spectral empirical BRDFs [58]—as there is often significant uncertainty about object material properties due to the usual absence of pre-launch optical calibration data and degradation of

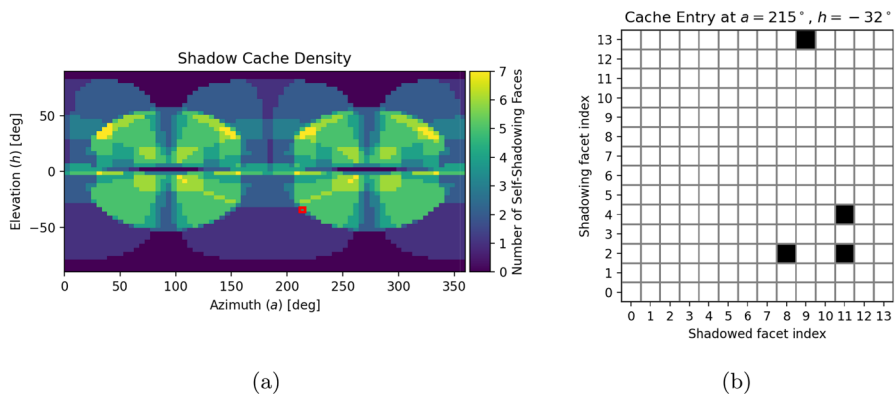


Fig. 4 Example of a self-shadow cache data structure **a** with $n_{Az} = 75$ and $n_{El} = 50$ built for the assumed box-wing satellite model shown in Fig. 5. In this example, the nearest cache entry **b** to the input location $Az = 215^\circ$, $El = -32^\circ$ shows that face 2 is shadowing both 8 and 11, while face 4 shadows 11 and face 13 shadows 9

materials on orbit [59]. Assuming a particular value of any unknown variables, e.g., the roughness of an observed solar panel, enforces an unnecessarily detailed prior on the inversion algorithm. This is not to say that a simpler BRDF permits more accurate material assumptions, but merely that it is more responsible to keep models simple under high uncertainty. In exchange, using a simple BRDF model permits more initial condition optimizations in a given amount of CPU time, leading to denser and more informative solution sets.

For this work, we assume prior knowledge of the reflective coefficients parameterizing the BRDF, namely the coefficient of diffuse reflectivity C_d , coefficient of specular reflectivity C_s , and the specular exponent n . The overall surface reflectance for an illumination direction $\hat{s}$ and observation direction $\hat{o}$ is given in terms of the halfway vector $\hat{h} = (\hat{s} + \hat{o}) / \|\hat{s} + \hat{o}\|$ and the optical coefficients [60]:

$$f_r(\hat{s}, \hat{o}) = \frac{C_d}{\pi} + \frac{n+2}{2\pi} \frac{C_s(\hat{n} \cdot \hat{h})^n}{4(\hat{n} \cdot \hat{s})(\hat{n} \cdot \hat{o})}. \quad (30)$$

3.5 Simulating Light Curves

The optical cross section f_p of a space object with the normal vector of the i th face $\hat{n}_i$, face BRDF $f_{r,i}(\hat{s}, \hat{o})$, illuminated from $\hat{s}_k$ and observed at $\hat{o}_k$ in the body frame is given by [11]:

$$f_p(\hat{s}_k, \hat{o}_k) = \sum_{i=1}^n \bar{a}_i(\hat{s}_k, \hat{o}_k) f_{r,i}(\hat{s}_k, \hat{o}_k) (\hat{n}_i \cdot \hat{s}_k) (\hat{n}_i \cdot \hat{o}_k). \quad (31)$$

Here, $\bar{a}_i(\hat{s}_k, \hat{o}_k)$ is the unshadowed area of the i th face, which can be simplified for convex objects as [61]:

$$\bar{a}_i(\hat{s}, \hat{o}) = \begin{cases} a_i & \text{if } (\hat{o} \cdot \hat{n}_i) > 0 \text{ and } (\hat{s} \cdot \hat{n}_i) > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (32)$$

For non-convex objects, $\bar{a}_i$ computed via Eq. (24). The mean analog-to-digital unit (ADU) counts $\bar{S}_{\text{SO},k}$ recorded by an optical CCD or CMOS sensor due to the reflected optical cross section $f_p(\hat{s}_k, \hat{o}_k)$ and the geometry of the observation is [62]:

$$\bar{S}_{\text{SO},k} = f_p(\hat{s}_k, \hat{o}_k) \frac{A \cdot \Delta t_k \cdot f_{\odot}(\mathbf{r}_k)}{g R_{\oplus}^2 r_k^2} \int_0^{\infty} P(\lambda) Q(\lambda) T_k(\lambda) I_{\odot}(\lambda) \left(\frac{\lambda}{hc} \right) d\lambda. \quad (33)$$

The integrand accounts for the quantum efficiency $Q(\lambda)$, the passband filter transmittance if one is used $P(\lambda)$, atmospheric transmission $T(\lambda)$, and the reflected irradiance spectrum of the object's materials—approximated as the irradiance spectrum of sunlight $I_{\odot}(\lambda)$. Integrating this quantity, scaled by the inverse energy of the photon with wavelength λ , $\lambda h^{-1} c^{-1}$, where h is Planck's constant in Joule-seconds and c is the speed of light in vacuum in meters per second, yields the fraction of reflected energy

absorbed into the sensor as photoelectrons. Accounting for the current distance from the center of the Earth to the Sun in AU $R_{\oplus}$, the distance from the observer to the object r_k in meters, the exposure time Δt in seconds, the fraction of solar irradiance $f_{\odot}(\mathbf{r}_k)$ reaching the object at ECI position $\mathbf{r}_k$ —accounting for the Earth’s shadow, the sensor gain g in ADU per photoelectron, and the effective telescope aperture area A in m^2 yields the mean ADU signal due to the object. $f_{\odot}(\mathbf{r}_k)$ may be computed through a simple penumbra model as described in [62], or a higher-fidelity approach accounting for Earth’s atmosphere as in [63] if precise penumbra entry timing is required.

The integral in Eq. (33) may appear to introduce additional computational complexity for the inversion optimization, but since the integral is not a function of the space object’s attitude, it can be computed ahead of the inversion and reused for each subsequent light curve evaluation. We use the LOWTRAN atmospheric model to compute the atmospheric transmission spectrum $T(\lambda)$ [64].

We make several implicit assumptions in the formulation of Eq. (33). We assume that the object reflects the same spectrum as the Sun, such that the BRDF is independent of the incident irradiance spectrum. In reality, the amount and character of material reflectivity varies with wavelength [65], and these properties may change over time due to the harsh radiation environment on orbit [59, 66]. We also assume that the object’s orbit is known to determine the distance r . We also assume that the object is in full sunlight for the observation period, which is reasonable as measurements known to be taken of a space object during penumbra can be removed from the light curve before inversion.

The observed brightness in apparent magnitudes under a given passband sensitivity $P(\lambda)$ with respect to the Vega zero-point due to the space object’s optical cross section $f_p(\hat{\mathbf{s}}_k, \hat{\mathbf{o}}_k)$ is given by:

$$m_k = -2.5 \log \left[\frac{f_p(\hat{\mathbf{s}}_k, \hat{\mathbf{o}}_k)}{R_{\oplus}^2 r_k^2} \int_0^{\infty} P(\lambda) I_{\odot}(\lambda) d\lambda \left(\int_0^{\infty} I_{\text{Vega}}(\lambda) P(\lambda) d\lambda \right)^{-1} \right], \quad (34)$$

where $I_{\odot}(\lambda)$ is the exoatmospheric irradiance spectrum of the Sun, $R_{\oplus}$ is the current distance from the center of the Earth to the Sun in AU, r_k is the distance from the observer to the object, and $I_{\text{Vega}}(\lambda)$ is the irradiance spectrum of Vega. Equation (34) implicitly assumes that the object reflects a scaled solar spectrum towards the observer. $I_{\odot}(\lambda)$ should be replaced with spectral reflectance information if any is available.

3.5.1 Light Curve Noise

There are four significant sources of noise in observed light curves: the celestial background, atmospheric turbulence, sensor noise, and shot noise [67]. Pattern noise, while a non-negligible source of error in both CCD and CMOS sensors, is not considered in this work as it can appear in a variety of forms and should be dealt with on a sensor-by-sensor basis [68].

The sensor noise comprises multiple independent random variables originating from different processes within the CCD or CMOS sensor. Each pixel well accumulates additional electrons during exposure time due to thermal noise [29]. This ‘dark current’ is well-approximated as a Poisson random variable with mean and variance λ_{dark} . The dark current accumulates with time and is expressed in units of ADU per pixel per second. At readout, the pixel well charge values are shifted to ensure no pixels record negative charges. Together with the noise introduced by the analog-to-digital converter, this read noise is well-modeled as a zero-mean Gaussian distribution with standard deviation σ_{read} per pixel in ADU. The total standard deviation of the sensor noise over n_{pix} pixels is [29]:

$$\sigma_{\text{sensor}} = \sqrt{n_{\text{pix}} (\Delta t \cdot \lambda_{\text{dark}} + \sigma_{\text{read}}^2)}. \quad (35)$$

Small differences in quantum efficiency between pixels of the sensor produces a so-called flat field effect, which is well-modeled as a zero-mean Gaussian with standard deviation σ_{flat} that scales linearly with the signal in each pixel via a constant f_k [69]. Across n_{pix} pixels each with a variable signal value S_i in ADU, the total standard deviation in ADU is:

$$\sigma_{\text{flat}} = f_k \sqrt{\sum_{i=1}^{n_{\text{pix}}} S_i}. \quad (36)$$

If light curve inversion is being performed with real data, the celestial background and sensor noise are observed as the standard deviation of the image background not containing object or star signals. In this work, we use synthetic data, simulating the variance of the celestial background through empirically derived models for the scattered moonlight, integrated starlight, twilight, zodiacal light, airglow, and light pollution [12]. These terms are all well-modeled as independent Poisson random variables, yielding an overall Poisson random variable with mean and variance λ_{back} across n_{pix} pixels:

$$\lambda_{\text{back},k} = n_{\text{pix},k} (\lambda_{\text{moon},k} + \lambda_{\text{star},k} + \lambda_{\text{twi},k} + \lambda_{\text{zod},k} + \lambda_{\text{air},k} + \lambda_{\text{poll},k}). \quad (37)$$

Atmospheric turbulence—or scintillation—at different altitudes in the atmosphere distorts and degrades all exoatmospheric signals that reach the telescope aperture [70]. We approximate the noise introduced by this effect on the overall photometric brightness measurement through Young’s approximation [71], expressed at timestep k as a function of the telescope aperture diameter D in meters, the exposure time Δt_k in seconds, the angle of the observation from zenith γ_k , the altitude of the observing station h_{obs} in meters, and the turbulence scaleheight $H \approx 8000$ meters. Young’s approximation for the normalized scintillation noise variance is [70]:

$$\sigma_{Y,k}^2 = 10^{-5} D^{-4/3} \Delta t_k^{-1} \cos^{-3}(\gamma_k) e^{-\frac{2h_{\text{obs}}}{H}}. \quad (38)$$

The remaining noise source is simply the Poisson process of arriving photons from the object—termed shot noise [29]. Given the mean space object signal in ADU computed in Eq. (33), the shot noise variance at timestep k is simply a function of the sensor gain g in electrons per ADU:

$$\lambda_{\text{shot},k} = \frac{\bar{S}_{\text{SO},k}}{\sqrt{g}}. \quad (39)$$

Taking all noise sources into account and assuming each is independent, the variance of the light curve at timestep k is given by:

$$\sigma_{S,k}^2 = \lambda_{\text{back},k} + \bar{S}_{\text{SO},k} \sigma_{Y,k}^2 + \lambda_{\text{shot},k} + \sigma_{\text{flat}}^2 + \sigma_{\text{sensor}}, \quad (40)$$

Such that the light curve is well-approximated by a time-varying Gaussian distribution:

$$S_{\text{SO},k} \sim \mathcal{N}(\bar{S}_{\text{SO},k}, \sigma_{S,k}^2). \quad (41)$$

4 Attitude Inversion Results with Varying Inertia Uncertainty

To evaluate the proposed attitude inversion algorithm, we simulated realistic ground-based light curve measurements of GEO objects and performed attitude inversion. Table 1 provides parameters for the Purdue Optical Ground Station telescope used to simulate the observations for all test cases. In addition to the values in Table 1, the quantum efficiency curve $Q(\lambda)$ for the Purdue Optical Ground Station may be found in the sensor datasheet [72], and the solar irradiance spectrum $I_{\odot}(\lambda)$ required for Eq. (33) matches that used by the European Space Agency’s PROOF tool [73].

The truth initial conditions for the object attitude profiles listed in Table 2 are held constant between all test cases. The initial angular velocity was selected specifically to promote complex, tumbling attitude profiles to push the limits of the solver. We choose torque-free dynamics with $\mathbf{M} = \mathbf{0}$ in Eq. (11) as the light curves used in this work are short—on the order of minutes—meaning that environmental torques due to solar radiation pressure, drag, the gravity gradient, and eddy currents are negligible. This implicitly assumes that the observed object is not thrusting during observation.

Table 1 Telescope information common to all test cases

Variable	Value
Effective aperture area A	0.076 m ²
Observer location	32.900° N, − 105.533° W 2.24 km above MSL
Pixels containing source n_{pix}	100
Sensor read noise σ_{read}	9 ADU/pixel
Sensor dark current rate λ_{dark}	0.01 ADU/pixel/s
Sensor flat field strength f_k	0.01 nondim
Sensor gain g	5.1 e [−] /ADU

Table 2 True object attitude values for all test cases

Variable	Value
Initial orientation $p(0)$	$-\frac{1}{3} [1 \quad 1 \quad 1]^T [\text{nondim}]$
Initial angular velocity $\omega(0)$	$[0.1 \quad 0.1 \quad 0.1]^T [\text{rad/s}]$
External torque M	$[0.0 \quad 0.0 \quad 0.0]^T [\text{N m}]$

Table 3 Inversion solver configuration for all test cases

Variable	Value
BFGS maximum function evaluations	1000
BFGS maximum iterations	100
BFGS finite difference step size	$1 \cdot 10^{-3}$
BFGS gradient ∞ -norm tolerance	$1 \cdot 10^{-5}$
Shadow cache size $n_{Az} = n_{EI}$	200

Table 4 Reflection properties of real materials. The authors estimate the MLI and white paint parameters qualitatively

Material	C_d	C_s	n
Solar panel [75]	0.15	0.25	0.26
Aluminum bus structure [75]	0.34	0.40	8.9
Multilayer insulation (MLI)	0.1	0.9	20
White paint	0.9	0.1	1

The initial state guesses for the optimization for each case are sampled using the procedure defined in Sect. 2.3 with $\omega_{\max} = 1.5\omega_{\text{true}}(0)$. This scaling factor of 1.5 was selected to disperse initial guesses widely about the truth, simulating real-world uncertainty in the angular velocity magnitude. When working with real data, this bound may be chosen via a periodogram or the dominant frequency in the observed signal as described in [74]. The configuration of the BFGS solver used for all test cases is listed in Table 3.

The 3D models in all test cases share the same set of materials, listed in Table 4. The coefficients in Table 4 were originally developed for the Phong BRDF and are used interchangeably with the Blinn-Phong BRDF in this work. These properties are assumed to be known for the purposes of our investigation.

We present four results cases to evaluate the proposed attitude inversion algorithm. The first two cases are for a rocket body, while the second two cases observe a box-wing satellite. The first case for each object is a situation of low prior inertia uncertainty, e.g., attitude inversion in cooperation with satellite operators. The second case for each object has much higher inertia uncertainty, e.g., attitude inversion of an object with limited available documentation. Relevant object parameters that change in each case are listed in Table 5. Cases for the same object share the same light curve to isolate the effect of the varying inertia ratio uncertainty. Particular observation start time, integration time, and observation frequency values are held constant for the same object but vary between objects to produce measurements with a moderate noise level. All results presented in this section were produced using an Apple M1 CPU.

In all test cases, the measured light curve brightness values along with the observation geometry are assumed to be known. Light curves are simulated with a true

Table 5 Overview of major varying parameters in results cases including the true inertia tensor uncertainties J_y/J_x and J_z/J_x , as well as the prior uncertainty in the object's inertia ratios σ_J

Case	True (J_y/J_x , J_z/J_x)	Inertia ratio uncertainty σ_J	Object CO-SPAR ID
1a	(1.0, 0.5)	0.1	1987-084D
1b	(1.0, 0.5)	1.0	1987-084D
2a	(0.345, 1.034)	0.1	2014-004A
2b	(0.345, 1.034)	1.0	2014-004A

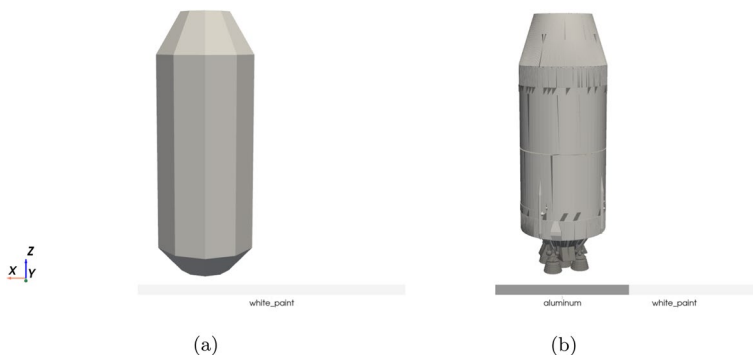


Fig. 5 The assumed **a** and measured **b** models used in Cases 1a and 1b with color-coded materials. Measured model sourced from NASA's 3D model archive [76]

object model while the attitude inversion is performed using a simplified shape representation. This model mismatch improves the realism of the results as such discrepancies occur frequently in real data. The orientation and angular velocity of the object are explicitly unknown.

4.1 Cases 1a and 1b: Rocket Body Attitude Inversion

Figure 5 displays the complex measured model and the simpler assumed model used for Cases 1a and 1b. The geometry of the models differs substantially in face count, convexity, and, to a lesser extent, material assignment.

Figure 6 displays the simulated light curves for the attitude inversion. Notably, most measurements are dominated by readout noise from Eq. (35), although some larger signal values are dominated by scintillation effects from Eq. (38). As the measured object is only slightly non-convex, computing its light curve with pixel-wise shadows produces similar results to the mean brightness values found for the assumed model, as shown in the lower right subplot. As the assumed model is convex, the choice of shadowing model has no impact on its predicted mean brightness. The model mismatch is apparent in the small deviations between the pixel-wise shadowed light curves of the assumed and measured models. The conversion from ADU to apparent magnitude for Fig. 6a is accomplished with Eqs. (33) and (34).

As noted in Table 5, the only parameter that changes between Cases 1a and 1b is the prior inertia ratio uncertainty σ_J . The observed light curve used in Cases 1a and 1b are the same and are produced by the parameters listed in Table 6.

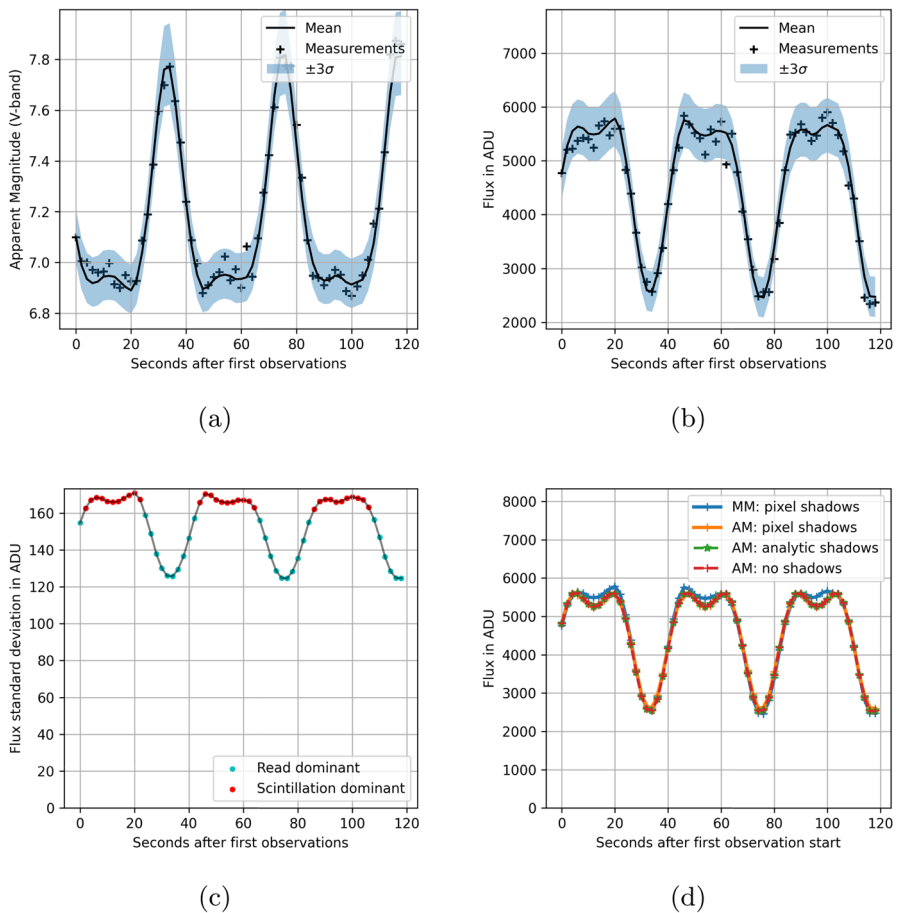


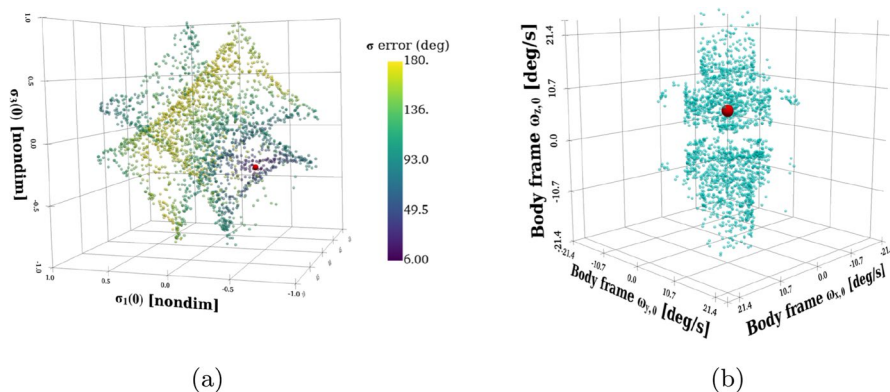
Fig. 6 Noisy light curves used for Cases 1a and 1b: **a** light curves in terms of apparent magnitude, **b** light curves in terms of ADU, **c** ADU measurement standard deviation, and **d** comparison of shadowing methods. In the shadowing methods comparison, AM and MM are the assumed and measured models, respectively

Table 6 Light curve simulation parameters for Cases 1a and 1b

Variable	Value
True inertia ratios $[J_y/J_x \quad J_z/J_x]$	[1.0 0.5]
Initial simulation date	October 10, 2024 08:00:00.000 UTC
Object identity (COSPAR ID)	1987-084D
Light curve duration	2 min
Measurement frequency	0.5 Hz
Integration time	0.01 s

Table 7 Solution statistics for Case 1a

Variable	Value
Ideal loss f_{best}	0.52
Loss bound α	2
Number of sampled initial conditions	10^5
Convergence time	9.8 min
Number of converged solutions	2357 (2.357%)
Number of overfit solutions	0
Best and median solution losses	75, 0.95
Median BFGS iterations	7
Median $\ x_k - x_0\ $	0.456
Median $\ \omega(0)\ $ error	21.34%
Median precession rate error	0.45%
Median spin rate error	−95.60%

**Fig. 7** Initial state solutions in MRP orientation space **a** and angular velocity space solutions **b** for Case 1, with true initial condition plotted in red

4.1.1 Case 1a: Rocket Body, Low Inertia Uncertainty

First, we study a case of attitude inversion of a rocket body with low uncertainty in its inertia tensor ratios. This scenario is a good baseline test for the efficiency and accuracy of our inversion scheme as the observed light curve is relatively simple, which, combined with the geometric symmetries of the object, should produce a wide variety of feasible solutions. Tables 5 and 6 provides simulation parameters for this test case, with solution statistics listed in Table 7.

Observing the positions that converged initial state solutions in orientation and angular velocity space reveals the geometry of the space of converged local minima. Figure 7a shows that the initial state MRP solutions lie on two disjoint tubes. The two families of orientation solutions correspond to reflecting the assumed cylinder about the phase angle bisector plane spanned by the vector halfway between the observer and Sun and their cross product—identical to the mediatrix plane identified in [33]. Figure 7b reveals that the angular velocity solutions lie mainly along four cylindrical sections symmetric about the body's axis of symmetry to maintain a constant preces-

sion rate. The solutions furthest from the origin in angular velocity space correspond to the small peak at twice the true precession rate.

Converged solutions selected with a loss constraint of $f_{\text{best}} < f_i < 2f_{\text{best}}$ are plotted in Fig. 8. The mean light curves produced by the converged solutions match the measurements well and appear to respect the changing measurement standard deviation as enforced by the negative log-likelihood loss function. This behavior is most evident around 10 s, where the uncertainty is the highest and the solutions are the most well-distributed, and 75 s, where the opposite is true. This behavior indicates that our choice of loss function is behaving as expected, yielding light curves whose variations follow the statistics of the measurements. Two families of angular velocity solutions expressed in body-frame azimuth and elevation are visible in Fig. 8d, separated by a band with no solutions surrounding zero elevation. The body-frame angular velocity directions are projected into azimuth and elevation components for display in a two-dimensional histogram. The azimuth and elevation are related to

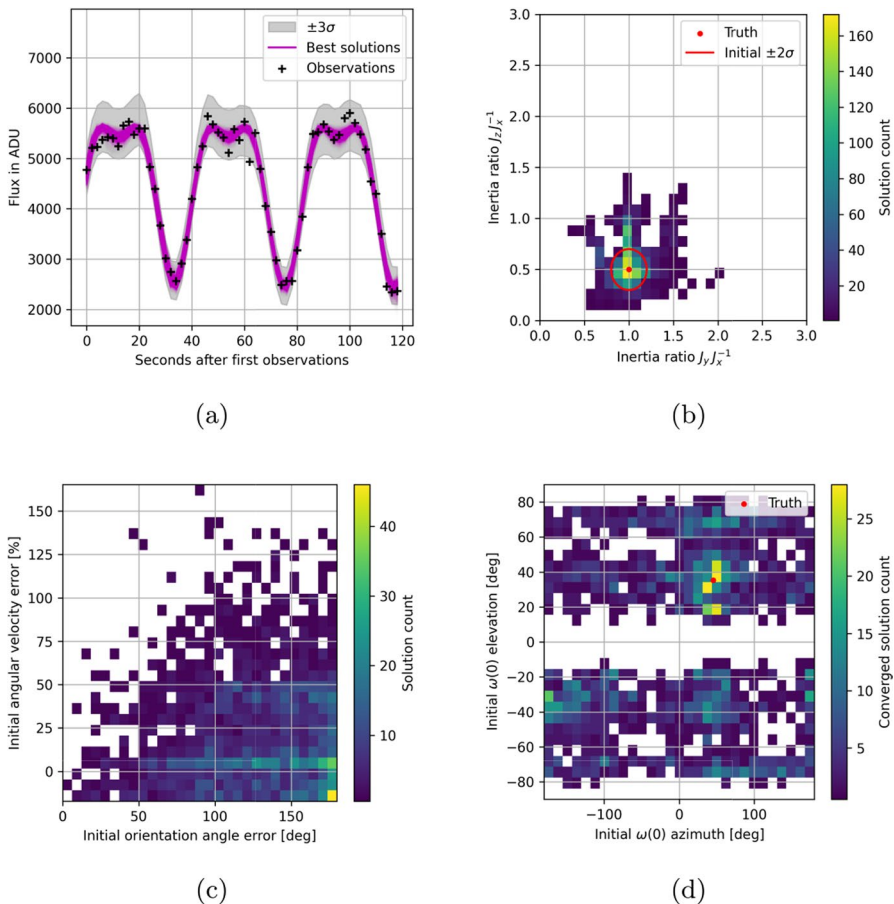


Fig. 8 Solutions to Case 1a: **a** Solution mean light curves displayed with the observations and $\pm 3\sigma$ bounds, **b** inertia ratio solutions, **c** initial orientation angle error—computed with Eq. (13)—and initial angular velocity magnitude error, and **d** initial angular velocity body frame azimuth and elevation

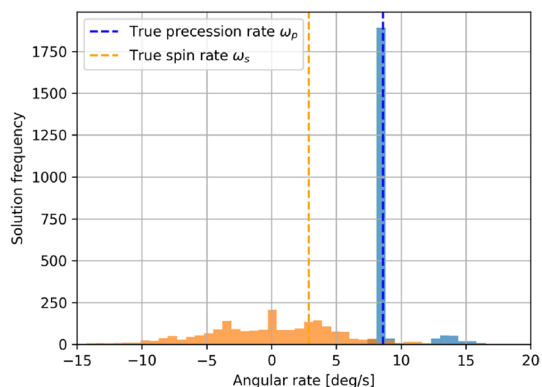
the Cartesian body frame via an inversion of Eq. (25), yielding $Az = \text{atan2}(y, x)$ and $El = \text{asin}(z/(x^2 + y^2 + z^2))$, where atan2 is the four-quadrant inverse tangent. These clusters are caused by the cylindrical symmetry of the assumed model—most angular velocity directions in these bands can produce a light curve with low error if given the correct initial orientation. Promisingly, the densest region of angular velocity direction lies around the truth.

The lower left reveals that most solutions fall near an orientation angle error of 180° . High angular errors are expected for an underdetermined inversion problem and are fully explained by the structure of the solution space revealed by Fig. 8a, where we see that the continuous surface of possible solutions traverses part of attitude space near a 180° offset from the truth. In general, the geometry of the S^3 hypersphere dictates that there are more orientations at a principal angle offset $\theta = 180^\circ$ than any other offset from a given orientation. This follows from the observation that the hypersphere equator perpendicular to a given orientation has a maximal area for a given relative θ , just as the equator of S^2 has the largest perimeter. Due to the construction of the quaternions in terms of trigonometric functions of $\theta/2$, that equator corresponds to a principal angle offset of 180° . The convergence of many particles near 180° merely displays the underdetermined nature of the problem, not maximum failure of the algorithm.

Finally, we note that the inertia ratios have converged well to the truth on the J_y/J_x axis but are more spread on J_z/J_x . This inertia ratio scaling is coupled with the high angular velocity magnitude errors—the inversion algorithm finds solutions that maintain the same precession rate while changing the spin rate. Since the spin rate of this axisymmetric object about its axis of symmetry is weakly observable in the light curve (at best), we cannot reliably determine the angular velocity magnitude or the axial moment of inertia. This observed ambiguity does not contradict well-accepted methods for angular velocity determination that use brightness ratios [77] or the periodogram [78], which do not attempt to estimate any axial rotation rates and assume the object's attitude is well-approximated as a single-axis spin.

Figure 9 supports our conclusion about the spin and precession rate observability in the light curve. We see that the majority of solutions converge well to the true precession rate, but the spin rate is almost entirely undetermined. This precession rate convergence is expected, as it solely determines the dominant behavior in the light

Fig. 9 Spin and precession rates for the converged solutions in Case 1a. The axial moment of inertia for this object is given by $J_a = J_z$ while the tangential term is approximated with $J_t = (J_x + J_y)/2 = (1 + J_y)/2$



curve. A small number of solutions appear at higher approximate precession rates due to a breakdown in the axisymmetric inertia tensor assumption when J_y is very different from $J_x = 1$. There are small local maxima in the spin rate distributions at the positive and negative true spin rates, likely resulting from a weakly observable variation in the light curve due to the engine and nozzle assembly which can be mimicked by the rotating flat faces of the simplified model.

4.1.2 Case 1b: Rocket Body, High Inertia Uncertainty

Figure 10 shows the converged solutions found for the prior inertia uncertainty $\sigma_J = 1.0$ and selection criteria $f_{\text{best}} < f_i < 2f_{\text{best}}$, with summary statistics in Table 8. Due to the higher initial inertia ratio uncertainty, fewer solutions are identified that meet the convergence bounds. The solutions that are found are qualitatively similar to

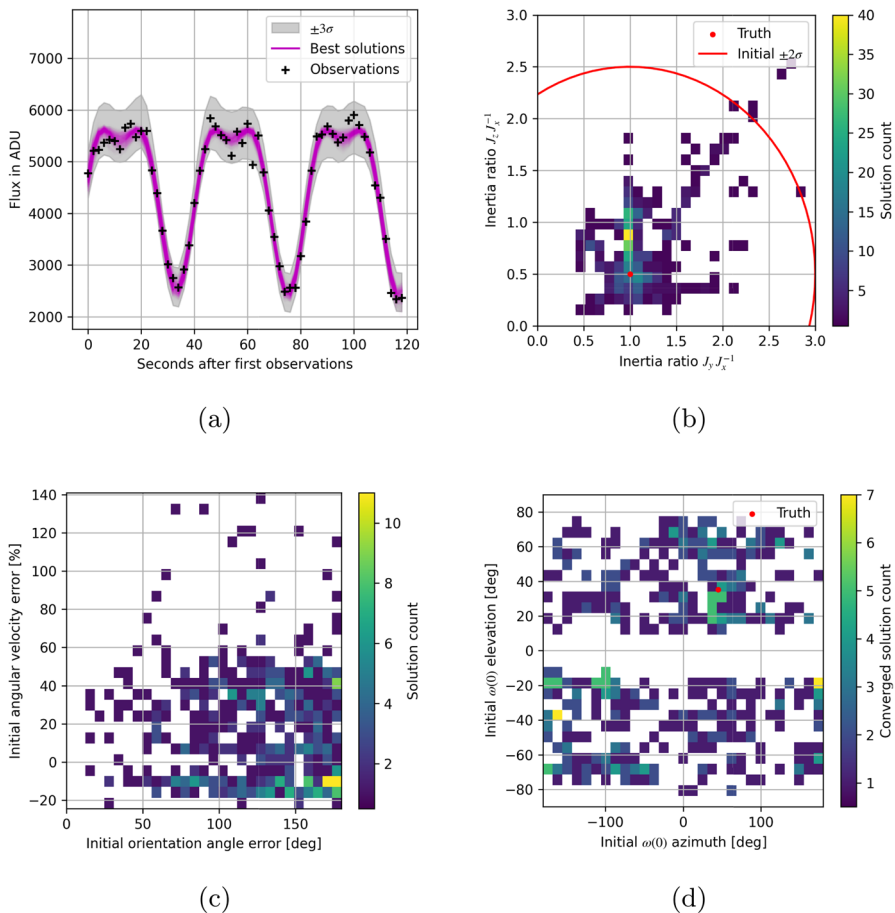


Fig. 10 Solutions to Case 1b: **a** Solution mean light curves displayed with the observations and $\pm 3\sigma$ bounds, **b** inertia ratio solutions, **c** initial orientation angle error—computed with Eq. (13)—and initial angular velocity magnitude error, and **d** initial state angular velocity body frame azimuth and elevation

Table 8 Solution statistics for Case 1b

Variable	Value
Ideal loss f_{best}	0.52
Loss bound α	2
Number of sampled initial conditions	10^5
Convergence time	8.8 min
Number of converged solutions	569 (0.569%)
Number of overfit solutions	0
Best and median solution losses	0.76, 0.96
Median BFGS iterations	7
Median $\ x_k - x_0\ $	0.419
Median $\ \omega(0)\ $ error	17.50%

those found in Case 1; they possess the same angular velocity bands, orientation, and angular velocity magnitude errors as seen in Table 8, although the inertia ratio solutions are slightly more widely distributed, producing more varied spin rate estimates than Case 1a while maintaining the correct precession rate. Taken together, Cases 1a and 1b provide evidence that for a tumbling cylinder, a larger inertia ratio uncertainty makes convergence more difficult, but the converged solutions are approximately the same quality.

4.2 Cases 2a and 2b: Box-Wing Attitude Inversion

The third and fourth cases use a highly non-convex box-wing satellite model. The measured model approximates HYLAS-4 (COSPAR 2018-033B, NORAD 43272), while the assumed model is a generic box-wing with a similar antenna configuration. Both models are displayed in Fig. 11. Compared to the first two cases, the models used for Cases 2a and 2b have significantly more model mismatch in the scale and geometry of their components. Figure 12 supports this conclusion, with the lower right subplot comparing the light curves produced by each model using pixel-wise shadowing, the semi-analytic shadowing presented in Sect. 3.3, and no shadowing. There are large deviations between the pixel-shadowed light curves of the respective models, most notably around 5 and 25 s.

The shadowing comparison also highlights two significant findings. First, the proposed semi-analytic shadowing produces nearly identical light curves to the reference pixel-wise shadowing, indicating that our new method correctly computes shadowed areas even for a maximum intersection recursion depth of $d = 2$. The no-shadowing light curve for the assumed model has 2.6 times higher average absolute error than the semi-analytic shadowing method compared to the reference pixel shadows. This particular attitude profile has only mild self-shadowing effects—the difference between the convex and self-shadowed light curves can be significantly larger for box-wings and other highly non-convex objects. The lower left subplot of Fig. 12 shows that the standard deviation of the measurements varies significantly over time and is dominated by the same noise terms seen in Fig. 6. The conversion from ADU to apparent magnitude for Fig. 12a is accomplished with Eqs. (33) and (34).

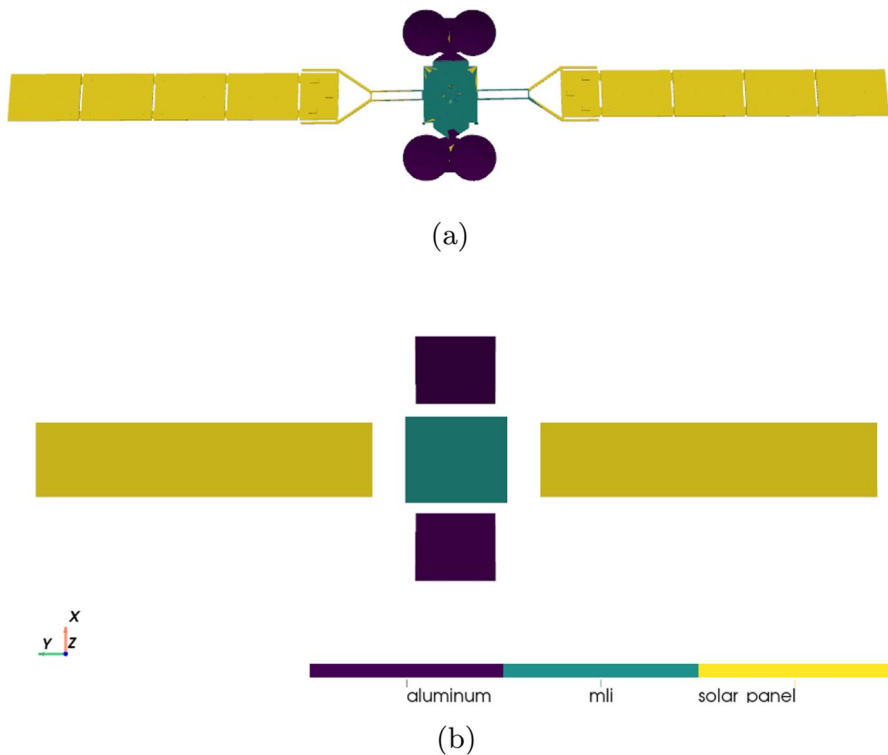


Fig. 11 The assumed **a** and measured **b** models used in Cases 2a and 2b with color-coded materials

As in Cases 1a and 1b, the only parameter that changes between Cases 2a and 2b is the prior inertia ratio uncertainty σ_J . The observed light curve is also held constant between Cases 2a and 2b and is produced by the parameters listed in Table 9.

4.2.1 Case 2a: Box-Wing, Low Inertia Uncertainty

The solutions from Case 2a are plotted in Fig. 13 with summary statistics provided in Table 10. The higher degree of model mismatch leads to fewer converged solutions, and several potentially overfit solutions. In this case, the loss bound α was lowered to 1.5 to improve overall solution quality at the cost of solution count. Even so, unlike in the first two cases, many solutions consistently exit the $\pm 3\sigma$ measurement bounds due to the model mismatch. Due to the increased complexity of the light curve and the geometry of the object, the solution space has more local minima, leading to faster BFGS convergence in fewer iterations and a smaller average distance $\|x_k - x_0\|$ between the initial guess and the nearest local minimum. Compared to Case 1, there is better convergence in the inertia tensor ratios as the object has three unique moments of inertia and no geometric axis of symmetry, removing the spin rate ambiguity seen in the rocket body cases. While this light curve does not uniquely determine the initial angular velocity direction, the angular velocity magnitude estimates have a median close to the truth as noted in Table 10. Likewise, initial orienta-

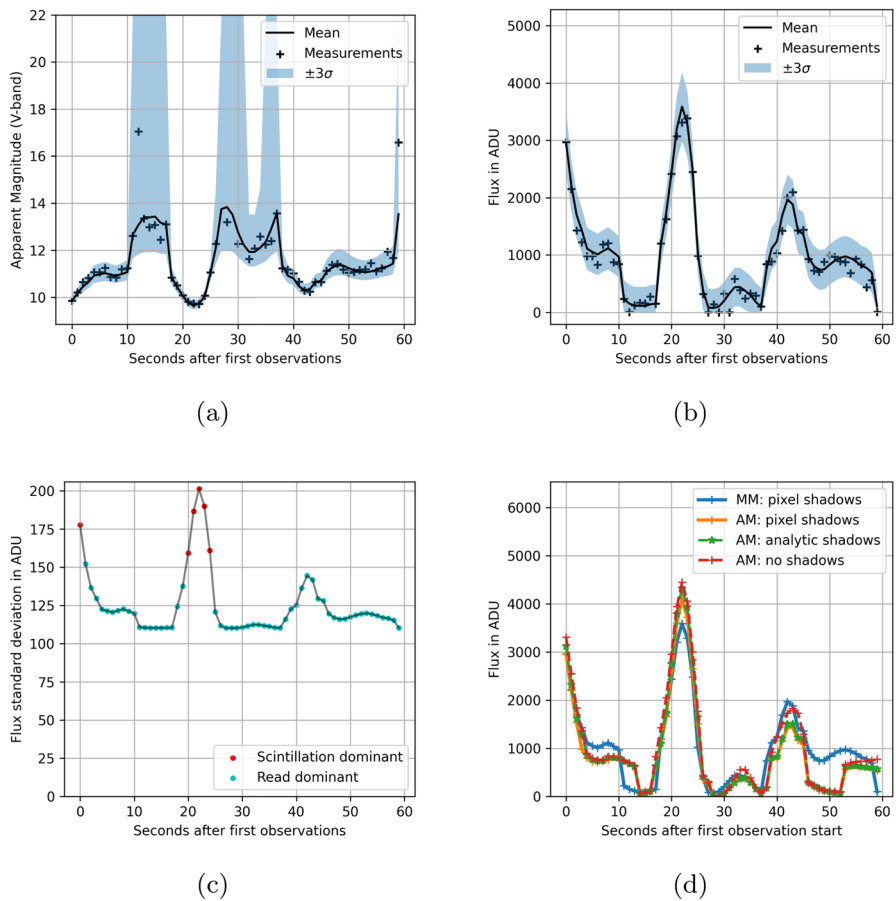


Fig. 12 Noisy light curves used for Cases 2a and 2b: **a** light curves in terms of apparent magnitude, **b** light curves in terms of ADU, **c** ADU measurement standard deviation, and **d** comparison of shadowing methods. In the shadowing methods comparison, AM and MM are the assumed and measured models, respectively

Table 9 Light curve simulation parameters for Cases 2a and 2b

Variable	Value
True inertia ratios [J_y/J_x J_z/J_x]	[0.345 1.034]
Initial simulation date	October 10, 2024 10:00:00.000 UTC
Object identity (COSPAR ID)	2014-004A
Light curve duration	1 min
Measurement frequency	1 Hz
Integration time	0.01 s

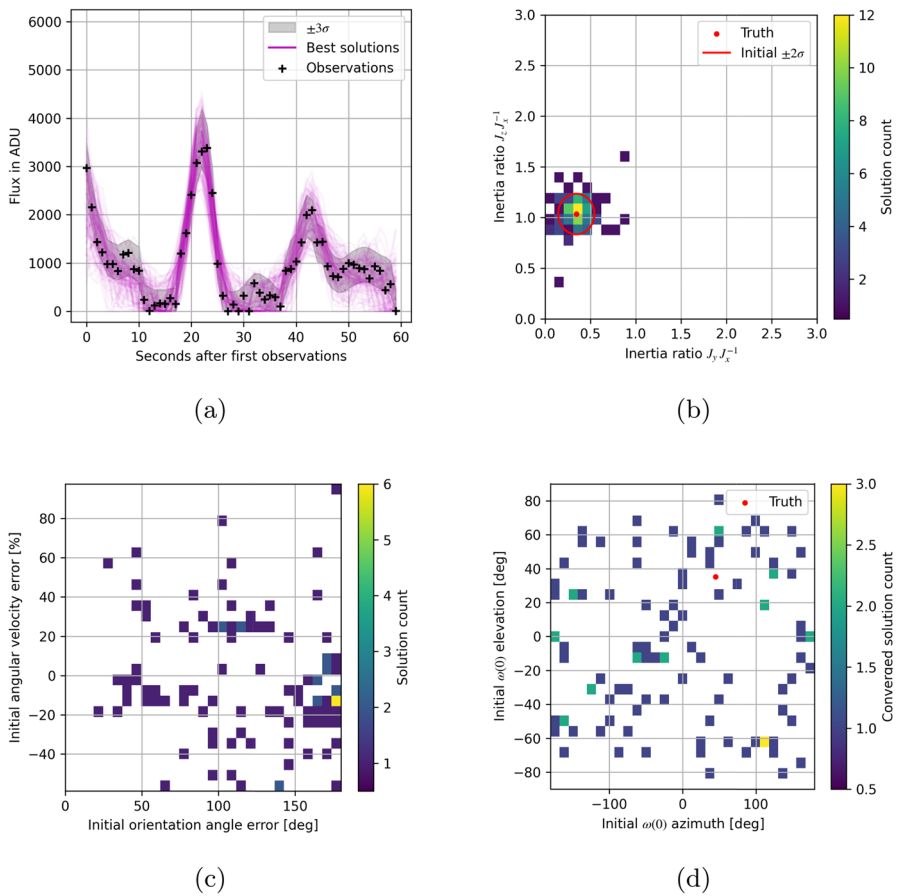


Fig. 13 Solutions to Case 2a: **a** Solution mean light curves displayed with the observations and $\pm 3\sigma$ bounds, **b** inertia ratio solutions, **c** initial orientation angle error—computed with Eq. (13)—and initial angular velocity magnitude error, and **d** initial state angular velocity body frame azimuth and elevation

Table 10 Solution statistics for Case 2a

Variable	Value
Ideal loss f_{best}	2.73
Loss bound α	1.5
Number of sampled initial conditions	10^5
Convergence time	8.0 min
Number of converged solutions	90 (0.090%)
Number of overfit solutions	28
Best and median solution losses	2.73, 3.71
Median BFGS iterations	4
Median $\ x_k - x_0\ $	0.360
Median $\ \omega(0)\ $ error	0.78%

tion errors of the solutions are high and show no apparent clustering. In conclusion, Case 2a highlights how quickly orientation and angular velocity direction solutions degrade when there is a significant model mismatch.

4.2.2 Case 2b: Box-Wing, High Inertia Uncertainty

The solutions from Case 2b are plotted in Fig. 14 with summary statistics provided in Table 11. The quality and number of the solution light curves in Case 2b are similar to those in Case 2a. As in Case 2a, the angular velocity solutions are widely distributed with a median error near zero, and orientation solutions possess large average errors. The only notable difference is the wide distribution of inertia ratio solutions owing to the higher uncertainty in their initial sampling. The results of Case 2b imply that if

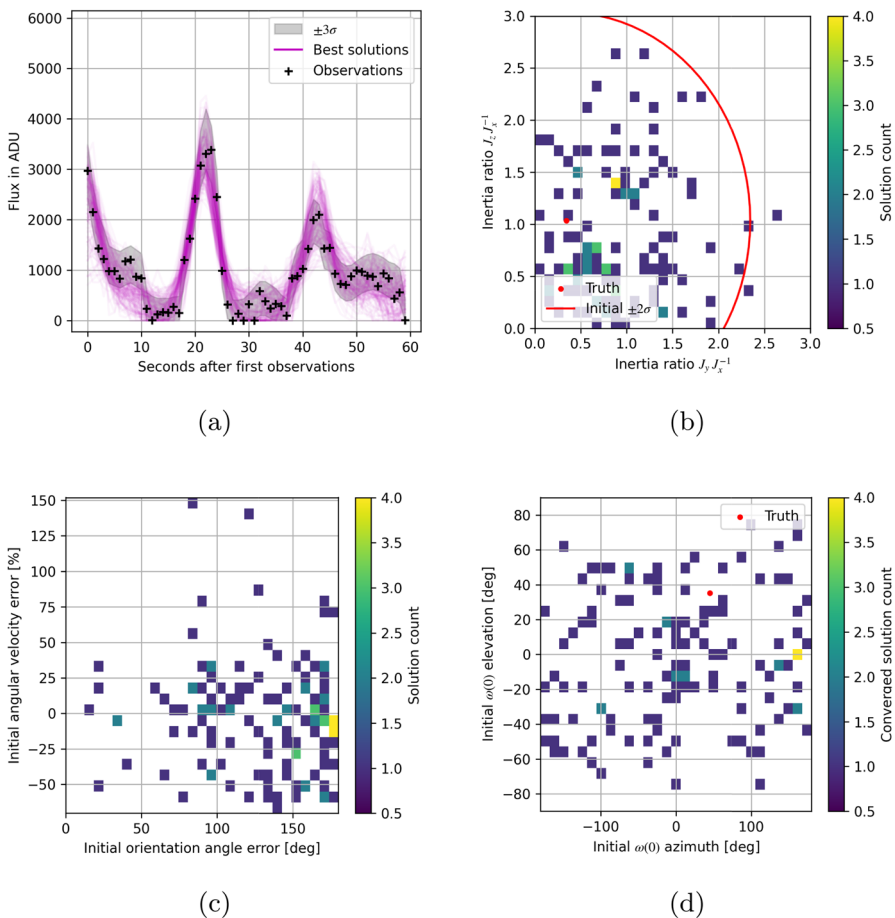


Fig. 14 Solutions to Case 2b: **a** Solution mean light curves displayed with the observations and $\pm 3\sigma$ bounds, **b** inertia ratio solutions, **c** initial orientation angle error—computed with Eq. (13)—and initial angular velocity magnitude error, and **d** initial state angular velocity body frame azimuth and elevation

Table 11 Solution statistics for Case 2b

Variable	Value
Ideal loss f_{best}	2.73
Loss bound α	1.5
Number of sampled initial conditions	10^5
Convergence time	8.3 min
Number of converged solutions	112 (0.005%)
Number of overfit solutions	16
Best and median solution losses	2.75, 3.67
Median BFGS iterations	4
Median $\ x_k - x_0\ $	0.384
Median $\ \omega(0)\ $ error	2.36%

there is a significant model mismatch and high inertia ratio uncertainty, converging any parameter besides the angular velocity magnitude is unlikely.

4.3 The Prospect of Merging Solutions

While it may be attractive to apply an averaging procedure for the initial orientation [79] or angular velocities to produce a single state vector, doing so will generally destroy the structure of the solution space spanned by the original solution set. Instead, we recommend that densely clustered regions of the solution space may be thinned without impacting this structure. Further pruning should be pursued only as new information becomes available. That information may be in the form of additional measurements which rule out some solutions, or refined object properties, either of which could also motivate the computation of additional sets of inversion solutions.

5 Conclusion

Attitude information informs many high-precision space situational awareness tasks including orbit propagation for high area-to-mass ratio objects and active debris removal. A space object's orientation is usually unresolved in ground-based optical imagery past LEO, but information can be recovered by tracking the time evolution of the object's brightness—the light curve. Estimating the orientation and angular velocity of a space object from its light curve is often under-determined even when the object's shape is known. Ambiguities may exist due to symmetries in the object's geometry, rotational inertia, observation geometry, and measurement noise. If no attitude information is available *a priori*, global optimization is needed.

We presented a new algorithm that addresses these realism shortcomings for light curve attitude estimation with noisy measurements and uncertainty in the ratios of the observed object's principal moments of inertia. Our improvements include a direct nonlinear optimization procedure that is easier to tune, a maximum likelihood loss function that discourages overfitting, and an semi-analytic shadowing method that enables fast light curve computation for highly non-convex space objects.

Results were shown for a rocket body in the first two cases using low and high inertia uncertainty, respectively, and the final two cases using low and high inertia uncertainty for a box-wing satellite.

Results for a rocket body in the first two cases show that dynamic and shape geometry symmetries can cause the spin rate to be unobservable, but the precession rate is well-estimated with $< 1\%$ error. The rocket body cases show that an increase in the inertia ratios uncertainty by a factor of 10 leads to a reduction in the number of converged solutions by a factor of ≈ 4 . That said, the converged light curves and the solution states are similar between the two inertia uncertainty levels, suggesting that inertia uncertainty does not dramatically change the nature of the inversion problem.

The final two cases study inversion for a box-wing and display a larger mismatch between the assumed object model and the measured object. This reduces the fraction of converged solutions by a factor of 5-25 compared to the first two cases and produces a small percentage ($< 0.05\%$) of solutions that overfit the light curve. Due to the box-wing's asymmetrical true inertia tensor, we find that the angular velocity magnitude is estimated with errors below 1% in the low inertia uncertainty case and 2.5% in the high uncertainty case. All test cases converge 10^5 initial guesses in under 10 min.

The presented results display the numerous ambiguities that can exist in the attitude inversion problem, making it generally difficult to guarantee that the best solution will be close to the truth. The best solutions in all four test cases do not converge to the true initial orientation, instead, they often have errors approaching 180° from the truth. Our results indicate that under realistic conditions of measurement noise and model mismatch, estimating the orientation of a space object uniquely from its light curve is unlikely. Even if attitude inversion indicates only one unique solution, the non-convex nature of the problem makes it difficult to determine if one or more better solutions were missed. If any one parameter does converge with high confidence—like the precession rate in the rocket body cases—the converged value could be applied as a constraint for a subsequent optimization run to narrow the search.

Logical extensions to this work would be applying these findings to real data and accounting for uncertainty in the optical properties of the object's materials.

Author Contributions L.R. developed the method, produced the results, and wrote the manuscript. C.F. supervised the work, provided feedback and guidance on the method and results, and commented on the manuscript.

Funding This work was partially supported by a National Defense Science and Engineering Graduate Fellowship.

Data Availability The optimization input and output produced in Sect. 4 for all test cases is available on request.

Declarations

Conflict of interest C.F. is a member of the editorial board.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long

as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Bonnal, C., Ruault, J.-M., Desjean, M.-C.: Active debris removal: recent progress and current trends. *Acta Astronaut.* **85**, 51–60 (2013). <https://doi.org/10.1016/j.actaastro.2012.11.009>
- Früh, C., Jah, M.K.: Coupled orbit attitude motion of high area-to-mass ratio (HAMR) objects including efficient self-shadowing. *Acta Astronaut.* **95**, 227–241 (2014). <https://doi.org/10.1016/j.actaastro.2013.11.017>
- Rachman, A.: Understanding Attitude Behavior of Inactive Glonass Satellites Using Spin Period Evolution. PhD Thesis, University of Bern (2023). <https://doi.org/10.48350/189927>
- Umansky, H., Clarke, K., Rogers, R., Hannon, A., Kelly, J., Latchu, T., Williams, A., Bryant, B.: Capstone: recovery and operations of a tumbling small satellite in deep space. In: *Proceedings of the Small Satellite Conference* (2023)
- Lemmens, S., Krag, H., Rosebrock, J., Carnelli, I.: Radar mappings for attitude analysis of objects in orbit. In: *Proceedings of the 6th European Space Debris Conference*, pp. 1–19 (2013)
- Park, T.H., D'Amico, S.: Adaptive neural-network-based unscented kalman filter for robust pose tracking of noncooperative spacecraft. *J. Guid. Control. Dyn.* **46**(9), 1671–1688 (2023). <https://doi.org/10.2514/1.G007387>
- Dennison, K.: Vision-Based Tracking and Shape Recovery of Non-cooperative Targets Using Distributed Space Systems. PhD Thesis, Stanford University (2023)
- Schildknecht, T., Hugentobler, U., Verdun, A.: Optical observations of space debris with the Zimmerwald 1-meter telescope. *Adv. Space Res.* **19**(2), 221–228 (1997). [https://doi.org/10.1016/S0273-1177\(97\)00004-5](https://doi.org/10.1016/S0273-1177(97)00004-5)
- Kobayashi, D., Frueh, C.: Image recovery for low earth orbit by leveraging turbulence and light curves. *J. Guid. Control. Dyn.* **47**(4), 623–637 (2024). <https://doi.org/10.2514/1.G007634>
- Burton, A.: Attitude Estimation Using Light Curves. PhD Thesis, Purdue University (2024). <https://doi.org/10.25394/PGS.26383684.v1>
- Fan, S.: The Light Curve Simulation and its Inversion Problem for Human-made Space Objects. PhD Thesis, Purdue University (2020). <https://doi.org/10.25394/PGS.12749570.v1>
- Robinson, L.: Light Curve Simulation and Shape Inversion for Human-Made Space Objects. Master's Thesis, Purdue University. <https://doi.org/10.25394/PGS.24728835> (2023)
- Vasile, M., Walker, L., Dunphy, R., Zabalza, J., Murray, P., Marshall, S., Savitski, V.: Intelligent characterisation of space objects with hyperspectral imaging. *Acta Astronaut.* **203**, 510–534 (2023). <https://doi.org/10.1016/j.actaastro.2022.11.039>
- Burton, A., Robinson, L., Frueh, C.: Light curve attitude estimation using particle swarm optimizers. *Adv. Space Res.* **74**(11), 5619–5638 (2024). <https://doi.org/10.1016/j.asr.2024.09.008>
- Tarantola, A.: Inverse Problem Theory and Methods for Model Parameter Estimation. Society for Industrial and Applied Mathematics, Philadelphia, PA (2005). <https://doi.org/10.1137/1.9780898717921>
- Fernández-Martínez, J.L., Niz, Z.F.-M., Pallero, J.L.G., Pedruelo-González, L.M.: From Bayes to Tarantola: new insights to understand uncertainty in inverse problems. *J. Appl. Geophys.* **98**, 62–72 (2013). <https://doi.org/10.1016/j.jappgeo.2013.07.005>
- Linares, R., Jah, M.K., Crassidis, J.L., Nebelecky, C.K.: Space object shape characterization and tracking using light curve and angles data. *J. Guid. Control. Dyn.* **37**(1), 13–25 (2014). <https://doi.org/10.2514/1.62986>
- Dianetti, A.D.: Resident Space Object Characterization Using Polarized and Multispectral Light Curves. PhD Thesis, University at Buffalo, The State University of New York. <https://www.proquest.com/dissertations-theses/resident-space-object-characterization-using/docview/238532188/se-2> (2020)

19. Burton, A., Frueh, C.: Two methods for light curve inversion for space object attitude determination. In: 8th European Conference on Space Debris (2021)
20. Gagnon, S.R.: Overcoming Convergence Issues with Attitude Determination from Light Curves. PhD thesis, University at Buffalo, The State University of New York. <https://www.proquest.com/dissertations-theses/overcoming-convergence-issues-with-attitude/docview/2926417092/se-2> (2024). Last updated 15 Feb (2024)
21. Wetterer, C.J., Jah, M.: Attitude determination from light curves. *J. Guid. Control. Dyn.* **32**(5), 1648–1651 (2009)
22. Linares, R., Crassidis, J., Jah, M.: Particle filtering light curve based attitude estimation for non-resolved space objects. *Adv. Astronaut. Sci.* **152**, 119–130 (2014)
23. Holzinger, M., Alfriend, K., Wetterer, C., Luu, K., Sabol, C., Hamada, K.: Photometric attitude estimation for agile space objects with shape uncertainty. *J. Gui. Control Dyn.* **37** (2014) <https://doi.org/10.2514/1.58002>
24. Vallverdú Cabrera, D., Utzmann, J., Förstner, R.: The adaptive Gaussian mixtures unscented Kalman filter for attitude determination using light curves. *Adv. Space Res.* **71**(6), 2609–2628 (2023). <https://doi.org/10.1016/j.asr.2022.11.027>
25. Clark, R., Dave, S., Wawrow, J., Lee, R.: Performance of parameterization algorithms for resident space object (RSO) attitude estimates. In: Proceedings of the 20th Advanced Maui Optical and Space Surveillance Technologies Conference, p. 14 (2019)
26. Piergentili, F., Santoni, F., Seitzer, P.: Attitude determination of orbiting objects from lightcurve measurements. *IEEE Trans. Aerosp. Electron. Syst.* **53**(1), 81–90 (2017). <https://doi.org/10.1109/TAES.2017.2649240>
27. Clark, R., Fu, Y., Dave, S., Lee, S.K.R.: Resident space object (RSO) attitude and optical property estimation from space-based light curves. *Adv. Space Res.* **70**(11), 3271–3280 (2022). <https://doi.org/10.1016/j.asr.2022.08.068>
28. Burton, A.B., Frueh, C., Robinson, L.: Attitude estimation using light curves: a particle swarm approach. In: AIAA SCITECH 2024 Forum (2024). <https://doi.org/10.2514/6.2024-0199>
29. Howell, S.B.: Handbook of CCD Astronomy, 2nd edn. Cambridge Observing Handbooks for Research Astronomers. Cambridge University Press, Cambridge, UK (2006)
30. Isoletta, G., Opromolla, R., Fasano, G.: Attitude motion classification of resident space objects using light curve spectral analysis. *Adv. Space Res.* **75**(1), 1077–1095 (2025). <https://doi.org/10.1016/j.asr.2024.10.034>
31. Linares, R., Crassidis, J.L.: Space object shape inversion via adaptive hamiltonian markov chain monte carlo. *J. Guid. Control. Dyn.* **41**(1), 47–58 (2018)
32. Gaylor, D., Anderson, J.: Use of hierarchical mixtures of experts to detect resident space object attitude. In: Advanced Maui Optical and Space Surveillance Technologies Conference, vol. 1, p. 70 (2014)
33. G. Marto, S., Vasile, M., Campbell, A., Murray, P., Marshall, S., Savitski, V.: Hyperspectral light-curve inversion for attitude determination. In: Proceedings of the AIAA SCITECH Forum (2024). <https://doi.org/10.2514/6.2024-0198>
34. Fan, S., Frueh, C.: A direct light curve inversion scheme in the presence of measurement noise. *J. Astronaut. Sci.* **67**(2), 740–761 (2019). <https://doi.org/10.1007/s40295-019-00190-3>
35. Robinson, L., Frueh, C.: Light Curve Inversion for Reliable Shape Reconstruction of Human-Made Space Objects. In: Proceedings of the 32nd AIAA/AAS Astrodynamics Specialist Conference, pp. 1–19 (2022)
36. Cabrera, D.V.: Characterization of Resident Space Objects Using Light Curves. PhD Thesis, Universität der Bundeswehr München (2024)
37. Marandi, S.R., Modi, V.J.: A preferred coordinate system and the associated orientation representation in attitude dynamics. *Acta Astronaut.* **15**(11), 833–843 (1987). [https://doi.org/10.1016/0094-5765\(87\)90038-5](https://doi.org/10.1016/0094-5765(87)90038-5)
38. Marsaglia, G.: Choosing a point from the surface of a sphere. *Ann. Math. Stat.* **43**(2), 645–646 (1972). <https://doi.org/10.1214/aoms/1177692644>
39. Shoemake, K.: III.6 - Uniform random rotations. In: Kirk, D. (ed.) Graphics Gems III (IBM Version), pp. 124–132. Morgan Kaufmann, San Francisco (1992). <https://doi.org/10.1016/B978-0-08-050755-2.50036-1>
40. Shuster, M.D.: Man, like these attitudes are totally random! I. Quaternions. In: Proceedings of the Astrodynamics Specialists Conference, vol. 108, pp. 383–396 (2001)
41. Shuster, M.D.: Survey of attitude representations. *J. Astronaut. Sci.* **41**(4), 439–517 (1993)

42. Broyden, C.G.: The convergence of a class of double-rank minimization algorithms 1. General considerations. *IMA J. Appl. Math.* **6**(1), 76–90 (1970). <https://doi.org/10.1093/imat/6.1.76>
43. Fletcher, R.: A new approach to variable metric algorithms. *Comput. J.* **13**(3), 317–322 (1970). <https://doi.org/10.1093/comjnl/13.3.317>
44. Goldfarb, D.: A family of variable-metric methods derived by variational means. *Math. Comput.* **24**, 23–26 (1970)
45. Shanno, D.F.: Conditioning of quasi-newton methods for function minimization. *Math. Comput.* **24**, 647–656 (1970)
46. Mohammad Nezhad, A., Aliakbari Shandiz, R., Eshraghniaie Jahromi, A.: A particle swarm-bfgs algorithm for nonlinear programming problems. *Comput. Op. Res.* **40**(4), 963–972 (2013). <https://doi.org/10.1016/j.cor.2012.11.008>
47. Wolfe, P.: Convergence conditions for ascent methods. *SIAM Rev.* **11**(2), 226–235 (1969). <https://doi.org/10.1137/1011036>
48. Hu, Y., Li, T.-M., Anderson, L., Ragan-Kelley, J., Durand, F.: Taichi: a language for high-performance computation on spatially sparse data structures. *ACM Trans. Graph.* **38**(6), 1–6 (2019). <https://doi.org/10.1145/3355089.3356506>
49. Butcher, J.C.: A history of Runge-Kutta methods. *Appl. Numer. Math.* **20**(3), 247–260 (1996)
50. Markley, F.L., Crassidis, J.L.: *Fundamentals of Spacecraft Attitude Determination and Control*. Springer, New York, NY (2014). <https://doi.org/10.1007/978-1-4939-0802-8>
51. Oguri, K.: Torque-free Attitude Motion: Axisymmetric Body. Course Notes, AAE 440/590: Spacecraft Attitude Dynamics, West Lafayette, IN. Course Notes
52. Group, K.: Calculating a Surface Normal. https://www.khronos.org/opengl/wiki/Calculating_a_Surface_Normal. Accessed 09 Oct 2024 (2013)
53. Allgower, E., Schmidt, P.: Computing volumes of polyhedra. *Math. Comput.* **46**(173), 171–174 (1986). <https://doi.org/10.2307/2008221>
54. Sutherland, I.E., Hodgman, G.W.: Reentrant polygon clipping. *Commun. ACM* **17**(1), 32–42 (1974). <https://doi.org/10.1145/360767.360802>
55. Blinn, J.F.: Models of light reflection for computer synthesized pictures. In: *Proceedings of the 4th Annual Conference on Computer Graphics and Interactive Techniques. SIGGRAPH '77*, pp. 192–198. Association for Computing Machinery, New York, NY, USA (1977). <https://doi.org/10.1145/563858.563893>
56. Cook, R.L., Torrance, K.E.: A reflectance model for computer graphics. *ACM Trans. Graph.* **1**(1), 7–24 (1982). <https://doi.org/10.1145/357290.357293>
57. Ashikhmin, M., Shirley, P.: An anisotropic Phong BRDF model. *J. Graph. Tools* **5**(2), 25–32 (2001). <https://doi.org/10.1080/10867651.2000.10487522>
58. Willison, A., Bedard, D.: Light curve simulation using spacecraft CAD models and empirical material spectral BRDFS. In: Ryan, S. (ed.) *Advanced Maui Optical and Space Surveillance Technologies Conference*, p. 101 (2015)
59. Keil, D., Sznajder, M., Renger, T., Bartels, N., Seefeldt, P., Riede, W., Spröwitz, T., Dekorsy, T.: Spacecraft materials' reflectivity and surface morphology: aging caused by proton irradiation. *J. Astronaut. Sci.* **70**(6), 50 (2023). <https://doi.org/10.1007/s40295-023-00418-3>
60. Duvenhage, B., Bouatouch, K., Kourie, D.: Numerical verification of bidirectional reflectance distribution functions for physical plausibility. In: *SAICSIT '13: Proceedings of the South African Institute for Computer Scientists and Information Technologists Conference*, pp. 200–208 (2013). <https://doi.org/10.1145/2513456.2513499>
61. Kaasalainen, M., Torppa, J.: Optimization methods for asteroid lightcurve inversion: I. Shape determination. *Icarus* **153**(1), 24–36 (2001). <https://doi.org/10.1006/icar.2001.6673>
62. Krag, H.: A Method for the Validation of Space Debris Models and for the Analysis and Planning of Radar and Optical Surveys. PhD thesis, Technische Universität Braunschweig (2003)
63. Li, Z., Ziebart, M., Bhattarai, S., Harrison, D.: A shadow function model based on perspective projection and atmospheric effect for satellites in eclipse. *Adv. Space Res.* **63**(3), 1347–1359 (2019). <https://doi.org/10.1016/j.asr.2018.10.027>
64. Kneizys, F.X., Shettle, E.P., Abreu, L.W., Chetwynd, J.H., Anderson, G.P.: Users Guide to Lowtran 7. Technical Report, Air Force Geophysics Lab (1988)

65. Najera, A., Velez-Reyes, M., Erives-Contreras, H., DeBlasio, D.: Simulating ground-based measurements of spectral signatures of resident space objects. In: Velez-Reyes, M., Messinger, D.W. (eds.) *Algorithms, Technologies, and Applications for Multispectral and Hyperspectral Imaging XXIX. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, vol. 12519, p. 1251908 (2023). <https://doi.org/10.1117/12.2665188>
66. Pearce, E.C., Weiner, B., Krantz, H.: Examining the effects of on-orbit aging of SL-12 rocket bodies using visible band spectra with the MMT telescope. *J. Space Safety Eng.* **7**(3), 376–380 (2020). <https://doi.org/10.1016/j.jsse.2020.07.017>.
67. Merline, W.J., Howell, S.B.: A realistic model for point-sources imaged on array detectors: the model and initial results. *Exp. Astron.* **6**(1–2), 163–210 (1995). <https://doi.org/10.1007/BF00421131>
68. Huang, S., Lu, T., Lu, Z., Rong, J., Zhao, X., Li, J.: Cmos image sensor fixed pattern noise calibration scheme based on digital filtering method. *Microelectron. J.* **124**, 105431D (2022). <https://doi.org/10.1016/j.mejo.2022.105431>
69. Newberry, M.: Pixel Response Effects on CCD Camera Gain Calibration. Technical Report, Mirametrics, Inc. (1996)
70. Osborn, J., Föhring, D., Dhillon, V.S., Wilson, R.W.: Atmospheric scintillation in astronomical photometry. *Mon. Not. R. Astron. Soc.* **452**(2), 1707–1716 (2015). <https://doi.org/10.1093/mnras/stv1400>
71. Young, A.T.: Photometric error analysis. VI. Confirmation of Reiger's theory of scintillation. *Astron. J.* **72**, 747 (1967). <https://doi.org/10.1086/110303>
72. Onsemi: 4096 (H) X 4096 (V) Full Frame CCD Image Sensor. Onsemi. Onsemi. Rev. 2 (2015)
73. Krag, H., Beltrami-Karlezi, P., Bendisch, J., Klinkrad, H., Rex, D., Rosebrock, J., Schildknecht, T.: Proof—The extension of Esa's master model to predict debris detections. *Acta Astronaut.* **47**(2), 687–697 (2000). [https://doi.org/10.1016/S0094-5765\(00\)00106-5](https://doi.org/10.1016/S0094-5765(00)00106-5).
74. Robinson, L., Frueh, C.: Optimal light curve attitude inversion with measurement noise: two case studies. In: *Proceedings of the 9th European Conference on Space Debris*, Bonn, Germany (2025). European Space Agency. <https://conference.sdo.esoc.esa.int/proceedings/sdc9/paper/245>
75. Fankhauser, F., Tyson, J.A., Askari, J.: Satellite optical brightness. *Astron. J.* **166**(2), 59 (2023). <https://doi.org/10.3847/1538-3881/ace047>
76. Keeter, W.: NASA 3D Models. <https://nasa3d.arc.nasa.gov/models/>. Accessed 27 Apr 2022 (2013)
77. Williams, V.: Location of the rotation axis of a tumbling cylindrical earth satellite by using visual observations: Part I: Theory. *Planet. Space Sci.* **27**(6), 885–890 (1979)
78. Šilha, J., Pittet, J.-N., Hamara, M., Schildknecht, T.: Apparent rotation properties of space debris extracted from photometric measurements. *Adv. Space Res.* **61**(3), 844–861 (2018). <https://doi.org/10.1016/j.asr.2017.10.048>
79. Cheng, Y., Markley, F.L., Crassidis, J.L., Oshman, Y.: Averaging quaternions. In: *Proceedings of the Astrodynamics Specialists Conference*. American Astronautical Society (2007)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.